

#1 · Professional Practice

What does it mean to take responsibility for your work and professional conduct as an MLA/T?

Answer

It means owning the accuracy and quality of every task you perform, admitting mistakes right away instead of hiding them, and following up to make sure errors are corrected. It also means being accountable to your supervisor, your patients, and the regulatory standards that govern lab work.

#2 · Professional Practice

Why must an MLA/T maintain confidentiality of all patient information?

Answer

Patient test results, personal identifiers, and health history are private and legally protected. Sharing this information with anyone who does not have a legitimate need to know it, even casually, breaks the law and damages patient trust in the lab.

#3 · Professional Practice

Scenario: A coworker asks you to share a patient's test result out of curiosity, not for care purposes. What should you do?

Answer

Decline to share the information. Confidentiality applies regardless of who is asking, and results should only be discussed with people who need them to provide or coordinate care. Report repeated requests like this to your supervisor if they continue.

#4 · Professional Practice

What does it mean to practice within your scope of competence?

Answer

It means only performing tasks you have been trained and authorized to do, and recognizing your own limits. When a task falls outside your training or experience, you must seek help from a supervisor or qualified colleague rather than attempting it alone.

#5 · Professional Practice

Scenario: You are asked to perform a collection technique you have never been trained on. What is the correct action?

Answer

Tell your supervisor or a qualified colleague that you have not been trained on this technique and ask for assistance or supervision. Attempting an unfamiliar procedure without help risks patient safety and goes against your scope of practice.

#6 · Professional Practice

What is the Health Care Consent Act, 1996?

Answer

It is Ontario legislation that sets the rules for obtaining valid consent before providing health care, including laboratory specimen collection. It confirms that a capable patient has the right to agree to or refuse a procedure after being given enough information to decide.

#7 · Professional Practice

What is the Canada Health Act?

Answer

It is federal legislation that establishes the criteria and conditions Canadian provinces and territories must meet to receive full federal funding for their health care systems. It underpins the public, publicly funded nature of Canadian health care, including laboratory services.

#8 · Professional Practice

What is the Regulated Health Professions Act (RHPA), 1991?

Answer

It is Ontario's umbrella law that governs regulated health professions, including how they are licensed and held accountable. Section 27 and its associated regulation, O. Reg. 107/96, define controlled acts and the exemptions that allow certain unregulated workers, such as MLA/Ts, to perform specific procedures like specimen collection.

#9 · Professional Practice

Under the RHPA, what does Section 27 and O. Reg. 107/96 cover?

Answer

Section 27 defines controlled acts, procedures that carry real risk of harm and normally can only be performed by authorized regulated professionals. O. Reg. 107/96 lists specific exemptions that permit MLA/Ts and other unregulated staff to legally perform certain controlled acts, such as blood collection, under defined conditions.

#10 · Professional Practice

What is the Laboratory and Specimen Collection Centre Act, Regulation 45/22?

Answer

It is Ontario legislation that sets the licensing and operating standards for medical laboratories and specimen collection centres. It covers requirements for facility operation, personnel, and quality to ensure labs provide safe and accurate testing services.

#11 · Professional Practice

What is PIPEDA?

Answer

PIPEDA stands for the Personal Information Protection and Electronic Documents Act. It is federal legislation that governs how private-sector organizations collect, use, and disclose personal information in the course of business, including some laboratory operations.

#12 · Professional Practice

What is the Mandatory Blood Testing Act, 2006?

Answer

It is Ontario legislation that allows a person exposed to another person's bodily fluids, such as through a needlestick injury, to apply for an order requiring that source person to undergo blood testing. It is relevant to MLA/Ts because of occupational exposure risk during specimen handling.

#13 · Professional Practice

What is PHIPA?

Answer

PHIPA stands for the Personal Health Information Protection Act. It is Ontario's health-specific privacy law that sets rules for how personal health information, including lab results, must be collected, used, disclosed, and protected by health information custodians.

#14 · Professional Practice

How does PHIPA differ in focus from PIPEDA?

Answer

PHIPA specifically governs personal health information within Ontario's health care system, while PIPEDA is a broader federal law covering personal information in commercial activities generally. Both can apply to a lab, but PHIPA is the more specific privacy rulebook for patient health data.

#15 · Professional Practice

What is the Ontario Human Rights Code?

Answer

It is provincial legislation that protects people from discrimination and harassment based on protected grounds such as race, sex, disability, and religion in areas including employment and services. It requires MLA/Ts to treat all patients and colleagues fairly and without bias.

#16 · Professional Practice

What is the Occupational Health and Safety Act (OHSA)?

Answer

It is Ontario legislation that sets out the rights and duties of workers and employers to protect health and safety in the workplace. In a lab setting, it covers things like the duty to report hazards, use protective equipment, and follow safe procedures.

#17 · Professional Practice

What is WHMIS?

Answer

WHMIS stands for the Workplace Hazardous Materials Information System. It is the Canadian standard for classifying and communicating information about hazardous materials in the workplace, using labels and safety data sheets so workers know how to handle chemicals safely.

#18 · Professional Practice

What is the Human Pathogens and Toxins Act?

Answer

It is federal legislation that regulates activities involving human pathogens and toxins to protect the health and safety of the public. It applies to labs that handle infectious specimens, setting requirements for safe handling, storage, and biosafety practices.

#19 · Professional Practice

What is the Transportation of Dangerous Goods Act (TDG)?

Answer

It is federal legislation and regulation that governs how dangerous goods, including biological and infectious substances such as blood specimens, must be classified, packaged, labelled, and transported safely between facilities.

#20 · Professional Practice

Scenario: A lab specimen needs to be shipped to an outside reference lab. What legislation governs how it must be packaged and labelled?

Answer

The Transportation of Dangerous Goods Act and its regulations apply, since biological specimens are classified as dangerous goods. The specimen must be packaged, labelled, and documented according to TDG requirements before it leaves the facility.

#21 · Professional Practice

Why must MLA/Ts use safe work practices at all times?

Answer

Lab work involves exposure to sharps, chemicals, and biological specimens that can carry infectious agents. Consistent use of safe practices, such as proper PPE and sharps disposal, protects the worker, coworkers, and patients from injury and infection.

#22 · Professional Practice

Why is it important to adhere to organizational policies and procedures?

Answer

Organizational policies translate legal and regulatory requirements into specific, practical steps for a given workplace. Following them consistently ensures quality, safety, and legal compliance, and protects both the patient and the MLA/T if something goes wrong.

#23 · Professional Practice

What is meant by the ethical principles and framework that influence medical laboratory practice?

Answer

These are the core values, such as respect for patient autonomy, doing good and avoiding harm, honesty, and fairness, that guide decision-making in situations not fully covered by law or policy. MLA/Ts are expected to recognize these principles and apply them in daily practice.

#24 · Professional Practice

Name common ethical principles that apply to health care practice.

Answer

Common principles include autonomy, respecting a patient's right to make their own decisions, beneficence, acting in the patient's best interest, non-maleficence, avoiding harm, and justice, treating people fairly. MLA/Ts use these to guide conduct in gray-area situations.

#25 · Professional Practice

What is the difference between direct supervision and general supervision for MLA/T and MLT practice?

Answer

Direct supervision means a supervisor is physically present and immediately available to oversee the task as it happens. General supervision means the supervisor is available for consultation, often by phone or nearby, but is not required to be physically present for every step.

#26 · Professional Practice

Scenario: An MLT is in the same room actively observing an MLA/T perform a task. What type of supervision is this?

Answer

This is direct supervision, since the supervisor is physically present and immediately available to intervene if needed. This level of oversight is typically required for higher-risk or less familiar procedures.

#27 · Professional Practice

Scenario: An MLT is reachable by phone from another department while an MLA/T performs a routine, familiar task alone. What type of supervision is this?

Answer

This is general supervision, since the supervisor is not physically present but remains available for guidance if a question or problem arises. This level is typically appropriate for routine tasks the MLA/T is competent and experienced in performing.

#28 · Professional Practice

Why should an MLA/T have knowledge of healthcare systems and professional/regulatory organizations?

Answer

Understanding how the healthcare system is structured, and knowing the role of bodies that regulate or represent the profession, helps an MLA/T understand where their role fits, who to turn to for guidance, and how standards of practice are set and enforced.

#29 · Professional Practice

What is the MLPAO?

Answer

The MLPAO is the Medical Laboratory Professionals' Association of Ontario, the professional association that represents MLA/Ts and MLTs in Ontario. It supports the profession through advocacy, education, and setting competency standards, such as the one this exam is based on.

#30 · Professional Practice

Why should an MLA/T understand the relationship between population health and the laboratory system?

Answer

Laboratory testing patterns are affected by trends in population health, such as disease outbreaks or chronic disease prevalence, and lab data in turn informs public health decisions. Understanding this connection helps an MLA/T appreciate the broader impact of accurate, timely lab work.

#31 · Professional Practice

Why is effective written and verbal communication a required competency for MLA/Ts?

Answer

Clear communication prevents errors in specimen labelling, patient identification, and result reporting, and ensures instructions from supervisors or other team members are understood correctly. Miscommunication in a lab setting can directly harm patient safety.

#32 · Professional Practice

What does interprofessional collaboration mean for an MLA/T?

Answer

It means working cooperatively with other members of the healthcare team, such as nurses, physicians, and other lab staff, sharing relevant information and coordinating tasks so that patient care runs smoothly and safely across disciplines.

#33 · Professional Practice

Give an example of interprofessional collaboration in a lab setting.

Answer

An MLA/T notifying a nurse promptly about a difficult blood draw so the care plan can be adjusted, or clarifying an unclear requisition with the ordering physician's office before proceeding, are both examples of collaborating across the healthcare team.

#34 · Professional Practice

Why must an MLA/T act with courtesy, consideration, and professionalism in all interactions?

Answer

Patients are often anxious or unwell during specimen collection, and colleagues rely on respectful teamwork to function well. Professional, courteous conduct builds trust, reduces patient distress, and reflects positively on the individual and the profession as a whole.

#35 · Professional Practice

What should an MLA/T do if they witness unprofessional behaviour from a colleague?

Answer

They should recognize the behaviour and report it to the appropriate leader according to their organization's policies. Staying silent allows unsafe or unethical conduct to continue, while proper reporting protects patients and upholds professional standards.

#36 · Professional Practice

What does it mean to respect the diversity and values of patients, colleagues, and other healthcare workers?

Answer

It means treating people fairly regardless of their culture, background, beliefs, or identity, and adapting your approach when needed to provide respectful, patient-centred care. This supports both ethical practice and compliance with human rights protections.

#37 · Professional Practice

Why are organizational and time management skills part of professional practice for an MLA/T?

Answer

Labs often handle high volumes of specimens under time pressure, and results can be time-sensitive for patient care. Good organization and time management help an MLA/T prioritize tasks correctly, avoid errors from rushing, and meet turnaround expectations.

#38 · Professional Practice

What is meant by performing troubleshooting and problem-solving within scope of practice?

Answer

It means being able to recognize and address routine issues, such as a minor equipment glitch or unclear labelling, using established procedures, while knowing when a problem is beyond your training and must be escalated to a supervisor or technologist.

#39 · Professional Practice

Why must an MLA/T be able to adapt and respond to changes in the workplace?

Answer

Lab procedures, technology, and policies change over time as standards and equipment evolve. An MLA/T who can adjust their practice smoothly helps maintain consistent quality and safety even as the work environment shifts.

#40 · Professional Practice

How can an MLA/T promote the image and status of the medical laboratory profession?

Answer

By maintaining high standards in their own work habits, presenting themselves professionally, and actively supporting the MLPAO. Individual conduct reflects on the whole profession's public reputation and credibility.

#41 · Professional Practice

Why does active support of the MLPAO matter for an individual MLA/T?

Answer

The MLPAO advocates for the profession, maintains competency standards, and provides resources for practitioners. Supporting it strengthens the profession's recognition and voice within the broader healthcare system, which benefits every practicing MLA/T.

#42 · Professional Practice

Why is continuous learning a required part of professional practice?

Answer

Lab techniques, technology, and regulations change over time, so maintaining and improving knowledge and skills ensures an MLA/T continues to provide safe, accurate, and current care. Continuous learning is an ongoing professional responsibility, not a one-time requirement.

#43 · Professional Practice

Scenario: A new piece of equipment is introduced in the lab and staff receive updated training. Which competency does keeping up with this training reflect?

Answer

This reflects continuous learning, maintaining and improving knowledge and skills over time. It also connects to adapting and responding to workplace changes, since new equipment often changes established routines.

#44 · Professional Practice

What is the general purpose of the group of laws listed under competency 1.4, taken together?

Answer

Together, these laws set the legal boundaries for consent, privacy, safety, and scope of practice that MLA/Ts must operate within. They cover everything from how patient information is protected to how hazardous materials and specimens must be handled and transported.

#45 · Professional Practice

Scenario: An MLA/T is unsure whether a task falls under their legal scope of practice due to controlled acts under the RHPA. What should they do?

Answer

They should stop and consult their supervisor or check the applicable exemption under O. Reg. 107/96 before proceeding. Practicing within scope means never assuming authorization; it must be confirmed against the actual legal exemptions that apply to MLA/Ts.

#46 · Medical Terminology Basic Biology Anatomy and Physiology

What is a prefix in medical terminology?

Answer

A prefix is a word part attached to the beginning of a root to change or add to its meaning, often showing location, number, or time. For example, hypo- means below or under and hyper- means above or excessive. Recognizing prefixes helps a lab worker quickly understand unfamiliar medical terms.

#47 · Medical Terminology Basic Biology Anatomy and Physiology

What is a suffix in medical terminology?

Answer

A suffix is a word part added to the end of a root that often changes the word into a noun, adjective, or describes a condition or procedure. Examples include -itis meaning inflammation and -ectomy meaning surgical removal. Suffixes are essential for reading requisitions and understanding why a test was ordered.

#48 · Medical Terminology Basic Biology Anatomy and Physiology

What does the root word 'hem/o' or 'hemat/o' mean?

Answer

Hem/o and hemat/o both refer to blood. This root appears in many lab terms such as hematology, the study of blood, and hemolysis, the breakdown of red blood cells. Recognizing this root helps identify blood-related tests and specimens quickly.

#49 · Medical Terminology Basic Biology Anatomy and Physiology

What does the suffix '-itis' mean, and give an example.

Answer

The suffix -itis means inflammation of a body part or organ. An example is hepatitis, which means inflammation of the liver. This suffix is common on requisitions and helps explain why certain blood tests were ordered.

#50 · Medical Terminology Basic Biology Anatomy and Physiology

What does the suffix '-emia' mean, and give an example.

Answer

The suffix -emia refers to a condition of the blood. An example is anemia, a condition where there is a low level of red blood cells or hemoglobin. This suffix commonly appears in lab result terminology.

#51 · Medical Terminology Basic Biology Anatomy and Physiology

What does the prefix 'brady-' mean, and how does it differ from 'tachy-'?

Answer

Brady- means slow, as in bradycardia, a slow heart rate. Tachy- means fast, as in tachycardia, a fast heart rate. These opposite prefixes are common on patient charts and help describe vital sign abnormalities.

#52 · Medical Terminology Basic Biology Anatomy and Physiology

What do the prefixes 'hypo-' and 'hyper-' mean?

Answer

Hypo- means below normal or deficient, while hyper- means above normal or excessive. For example, hypoglycemia is low blood sugar and hyperglycemia is high blood sugar. These prefixes are used constantly when interpreting lab values against reference ranges.

#53 · Medical Terminology Basic Biology Anatomy and Physiology

What does the root 'cyt/o' or the suffix '-cyte' mean, and give an example.

Answer

Both cyt/o and -cyte refer to a cell. An example is erythrocyte, meaning red blood cell. This root is central to hematology terminology because most cell types studied in the lab end in this suffix.

#54 · Medical Terminology Basic Biology Anatomy and Physiology

What are the four main body cavities described in MLA/T anatomy competencies?

Answer

The four main body cavities are the skull (cranial), thorax (chest), abdomen, and pelvis. Each cavity houses specific organs and structures that are protected by surrounding bone and tissue. Knowing these cavities helps a lab worker understand where organs are located relative to specimen collection sites.

#55 · Medical Terminology Basic Biology Anatomy and Physiology

Which body cavity contains the heart and lungs?

Answer

The thoracic cavity, located in the chest, contains the heart, lungs, and major blood vessels. It is protected by the rib cage and separated from the abdomen by the diaphragm. Understanding this location helps explain symptoms and test orders related to cardiac and respiratory conditions.

#56 · Medical Terminology Basic Biology Anatomy and Physiology

Which body cavity contains organs such as the stomach, liver, and kidneys?

Answer

The abdominal cavity contains organs such as the stomach, liver, spleen, and kidneys. It sits below the diaphragm and above the pelvis. Many abdominal organs are involved in digestion and metabolism, which relates directly to common chemistry lab tests.

#57 · Medical Terminology Basic Biology Anatomy and Physiology

Which body cavity houses the bladder and reproductive organs?

Answer

The pelvic cavity is located below the abdominal cavity and houses the bladder, rectum, and reproductive organs. It is protected by the bony pelvis. This cavity is relevant to urinalysis and reproductive hormone testing.

#58 · Medical Terminology Basic Biology Anatomy and Physiology

Why does the MLA/T need to know organ locations and their basic functions?

Answer

Knowing where organs are located and what they do helps a lab worker understand why a physician ordered a particular test and what an abnormal result might mean. For example, knowing the liver processes bilirubin helps explain a liver function panel. This knowledge supports safe, accurate specimen handling and patient communication.

#59 · Medical Terminology Basic Biology Anatomy and Physiology

Name the four main elements found in the human body that relate to proteins, carbohydrates, and fats.

Answer

The four main elements are hydrogen, oxygen, carbon, and nitrogen. Carbon, hydrogen, and oxygen form the backbone of carbohydrates and fats, while nitrogen is a key component of proteins. These elements combine in different arrangements to build the major molecules the body needs to function.

#60 · Medical Terminology Basic Biology Anatomy and Physiology

Which of the four main elements is unique to proteins compared to carbohydrates and fats?

Answer

Nitrogen is the element that distinguishes proteins from carbohydrates and fats. Proteins are built from amino acids, which each contain a nitrogen-based amino group. This is why nitrogen balance is often used clinically to assess protein metabolism.

#61 · Medical Terminology Basic Biology Anatomy and Physiology

What elements make up carbohydrates and fats?

Answer

Carbohydrates and fats are made primarily of carbon, hydrogen, and oxygen. Carbohydrates provide quick energy while fats store energy long term and support cell membrane structure. Both molecule types are broken down and measured in common chemistry panels.

#62 · Medical Terminology Basic Biology Anatomy and Physiology

Why is it useful for an MLA/T to understand basic biomolecule chemistry?

Answer

Understanding how proteins, carbohydrates, and fats are built from basic elements helps explain what common chemistry tests, such as glucose, cholesterol, and total protein, are actually measuring. This background supports accurate specimen processing and better understanding of abnormal results. It connects basic biology to everyday laboratory testing.

#63 · Medical Terminology Basic Biology Anatomy and Physiology

What is the human cell, in basic terms?

Answer

The human cell is the smallest structural and functional unit of the body, containing structures such as the nucleus, cytoplasm, and cell membrane. Cells group together to form tissues, which then form organs. Understanding basic cell structure supports interpreting cellular components seen in blood and other specimens.

#64 · Medical Terminology Basic Biology Anatomy and Physiology

What is epithelial tissue, and what is its main function?

Answer

Epithelial tissue covers body surfaces and lines organs and cavities, such as the skin and the lining of the digestive tract. Its main functions are protection, absorption, and secretion. This tissue type is often sampled in biopsies and cytology specimens.

#65 · Medical Terminology Basic Biology Anatomy and Physiology

What is connective tissue, and what is its main function?

Answer

Connective tissue supports, binds, and protects other tissues and organs in the body, and includes bone, cartilage, blood, and fat. It provides structural framework and helps transport substances throughout the body. Blood itself is classified as a connective tissue, which is directly relevant to laboratory work.

#66 · Medical Terminology Basic Biology Anatomy and Physiology

What is muscular tissue, and what is its main function?

Answer

Muscular tissue is responsible for movement through its ability to contract and relax. It includes skeletal muscle for voluntary movement, cardiac muscle for the heart, and smooth muscle in organs such as the intestines. Muscle activity relates to lab tests such as creatine kinase used to assess muscle damage.

#67 · Medical Terminology Basic Biology Anatomy and Physiology

What is nervous tissue, and what is its main function?

Answer

Nervous tissue is made up of cells such as neurons that transmit electrical signals throughout the body, allowing communication between the brain, spinal cord, and other organs. It controls and coordinates body functions and responses to stimuli. This tissue type is central to the nervous system's role in regulating the whole body.

#68 · Medical Terminology Basic Biology Anatomy and Physiology

What is DNA and what is its basic function in the cell?

Answer

DNA, or deoxyribonucleic acid, is the molecule that stores an organism's genetic instructions in the nucleus of the cell. It directs the production of proteins and is passed from cell to cell during replication. DNA structure and stability are central to genetic testing performed in the laboratory.

#69 · Medical Terminology Basic Biology Anatomy and Physiology

What is RNA and how does its role differ from DNA?

Answer

RNA, or ribonucleic acid, helps carry out the instructions stored in DNA by assisting in building proteins, and it can also be the genetic material itself in some viruses. Unlike DNA, RNA is typically single stranded and less stable. This difference is important when handling specimens for molecular testing, since RNA degrades more easily than DNA.

#70 · Medical Terminology Basic Biology Anatomy and Physiology

What is polymerase chain reaction (PCR) testing used for?

Answer

PCR is a laboratory technique that makes many copies of a specific piece of DNA so that even a very small amount of genetic material can be detected and analyzed. RNA targets, such as an RNA virus, must first be converted to DNA through reverse transcription (RT-PCR) before PCR can amplify them. It is widely used to identify infections, such as viral or bacterial pathogens, from patient specimens.

#71 · Medical Terminology Basic Biology Anatomy and Physiology

Why does DNA replication matter to understanding PCR?

Answer

DNA replication is the natural process where a cell copies its DNA before dividing, and PCR mimics this process in a controlled lab setting to amplify a target DNA sequence. Understanding how replication works, including the role of enzymes that build new DNA strands, helps explain how PCR can rapidly multiply genetic material for testing. This connection between natural biology and lab technique is a key concept for MLA/T certification.

#72 · Medical Terminology Basic Biology Anatomy and Physiology

What is the main function of the circulatory system?

Answer

The circulatory system transports blood, oxygen, nutrients, and waste products throughout the body using the heart, blood vessels, and blood. It plays a central role in maintaining homeostasis and delivering substances cells need to survive. Many common lab tests, such as complete blood counts, directly assess this system.

#73 · Medical Terminology Basic Biology Anatomy and Physiology

What is the main function of the digestive system?

Answer

The digestive system breaks down food into nutrients that the body can absorb and use for energy, growth, and repair, while eliminating waste. It includes organs such as the stomach, intestines, liver, and pancreas. Digestive function is commonly assessed through tests such as liver panels and stool analysis.

#74 · Medical Terminology Basic Biology Anatomy and Physiology

What is the main function of the endocrine system?

Answer

The endocrine system uses glands to produce and release hormones that regulate body processes such as metabolism, growth, and reproduction. Key glands include the thyroid, pancreas, and adrenal glands. Hormone levels, such as thyroid stimulating hormone, are frequently measured in the lab to assess this system's function.

#75 · Medical Terminology Basic Biology Anatomy and Physiology

What is the main function of the integumentary system?

Answer

The integumentary system consists of the skin, hair, and nails, and its main function is to protect the body from injury, infection, and fluid loss while helping regulate temperature. It is the body's first line of defense against the outside environment. This system is relevant to skin puncture specimen collection sites.

#76 · Medical Terminology Basic Biology Anatomy and Physiology

What is the main function of the immune and lymphatic system?

Answer

The immune and lymphatic system defends the body against infection and disease while also draining excess fluid from tissues back into the bloodstream. It includes structures such as lymph nodes, the spleen, and white blood cells. Lab tests such as white blood cell counts and differential counts directly assess this system.

#77 · Medical Terminology Basic Biology Anatomy and Physiology

What is the main function of the musculoskeletal system?

Answer

The musculoskeletal system provides structure, support, and movement for the body through bones, muscles, joints, and connective tissue. It also protects internal organs and stores minerals such as calcium. Tests related to this system include calcium levels and markers of muscle or bone damage.

#78 · Medical Terminology Basic Biology Anatomy and Physiology

What is the main function of the nervous system?

Answer

The nervous system controls and coordinates body activities by transmitting electrical signals between the brain, spinal cord, and the rest of the body. It regulates everything from voluntary movement to automatic functions like breathing and heart rate. Cerebrospinal fluid analysis is one laboratory test directly connected to this system.

#79 · Medical Terminology Basic Biology Anatomy and Physiology

What is the main function of the renal or urinary system?

Answer

The renal or urinary system filters waste products and excess fluid from the blood to form urine, helping maintain fluid and electrolyte balance in the body. Key organs include the kidneys, ureters, bladder, and urethra. Urinalysis and kidney function tests such as creatinine directly evaluate this system.

#80 · Medical Terminology Basic Biology Anatomy and Physiology

What is the main function of the reproductive system?

Answer

The reproductive system is responsible for producing offspring and includes organs that differ between males and females, along with hormones that regulate reproductive function. It also plays a role in producing sex hormones that affect other body systems. Hormone panels and pregnancy tests are common laboratory assessments related to this system.

#81 · Medical Terminology Basic Biology Anatomy and Physiology

What is the main function of the respiratory system?

Answer

The respiratory system is responsible for bringing oxygen into the body and removing carbon dioxide through breathing, using the lungs and airways. It works closely with the circulatory system to deliver oxygen to tissues throughout the body. Arterial blood gas testing is a key laboratory measure of respiratory function.

#82 · Medical Terminology Basic Biology Anatomy and Physiology

What is a common test used to monitor the circulatory system?

Answer

A complete blood count, or CBC, is a common test used to monitor the circulatory system by measuring red blood cells, white blood cells, and platelets. It can help detect conditions such as anemia or infection. This test is one of the most frequently ordered tests in a clinical laboratory.

#83 · Medical Terminology Basic Biology Anatomy and Physiology

What is a common test used to monitor the endocrine system?

Answer

Thyroid stimulating hormone testing is a common test used to monitor the endocrine system, particularly thyroid gland function. Abnormal levels can indicate an overactive or underactive thyroid. Hormone testing like this helps physicians diagnose and manage metabolic conditions.

#84 · Medical Terminology Basic Biology Anatomy and Physiology

What is a common test used to monitor the renal or urinary system?

Answer

A urinalysis is a common test used to monitor the renal or urinary system by examining the physical, chemical, and microscopic properties of urine. Blood creatinine levels are also commonly used to assess kidney filtering function. Together these tests help detect kidney disease or urinary tract problems.

#85 · Medical Terminology Basic Biology Anatomy and Physiology

Why is it important for the MLA/T to know which tests relate to which body systems?

Answer

Knowing which tests relate to which body systems helps a lab worker understand why a specific test was ordered and recognize when a result might be clinically significant. This knowledge supports accurate specimen collection, handling, and communication with the healthcare team. It connects the technical work of testing to the bigger picture of patient care.

#86 · Laboratory Mathematics Statistics and Quality Management

What is the SI base unit of volume used in the laboratory?

Answer

The litre (L) is the standard unit of volume, with millilitres (mL) and microlitres (μL) used for smaller amounts. Most lab instruments dispense and measure specimens in μL or mL. Using SI units consistently prevents dosing and dilution errors between departments.

#87 · Laboratory Mathematics Statistics and Quality Management

What is the SI unit of mass in the laboratory, and how is it usually subdivided?

Answer

The gram (g) is the base reference, broken down into milligrams (mg) and micrograms (μg) for small quantities like reagents or hormone levels. $1\text{ g} = 1000\text{ mg}$, and $1\text{ mg} = 1000\text{ }\mu\text{g}$. Reading the correct prefix matters because a missed decimal can change a result by a factor of 1000.

#88 · Laboratory Mathematics Statistics and Quality Management

Convert 98.6°F to Celsius.

Answer

Use $^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5/9$. That is $(98.6 - 32) \times 5/9 = 66.6 \times 5/9 = 37.0^{\circ}\text{C}$. Most lab refrigerators, freezers, and incubators are monitored in Celsius, so techs must be able to convert readings from older Fahrenheit equipment.

#89 · Laboratory Mathematics Statistics and Quality Management

What does 'linearity' mean when applied to a laboratory test?

Answer

Linearity is the range of concentrations over which an instrument's measured result stays directly proportional to the true concentration of the analyte. Outside that range the instrument reading becomes inaccurate, even though it still produces a number. Labs verify linearity with calibrators spanning the reportable range.

#90 · Laboratory Mathematics Statistics and Quality Management

Why does the laboratory use SI concentration units like mmol/L instead of older units like mg/dL?

Answer

SI units express concentration in a standardized, internationally comparable way based on moles rather than mass, which allows results to be compared across countries and literature. Canada primarily reports in SI units such as mmol/L and $\mu\text{mol/L}$. Consistent units reduce transcription and interpretation errors.

#91 · Laboratory Mathematics Statistics and Quality Management

Define the mean in statistics.

Answer

The mean is the arithmetic average, found by adding all values in a data set and dividing by the number of values. In quality control, the mean of repeated control runs represents the expected target value. A result that drifts away from the mean over time can signal a developing problem.

#92 · Laboratory Mathematics Statistics and Quality Management

Define median and mode, and how they differ from the mean.

Answer

The median is the middle value when data are arranged in order, and the mode is the value that occurs most often. Unlike the mean, the median is not distorted by one extremely high or low outlier. Mode is most useful for describing the most common category or result in a data set.

#93 · Laboratory Mathematics Statistics and Quality Management

What does standard deviation (SD) measure?

Answer

Standard deviation measures how spread out individual values are around the mean. A small SD means results cluster tightly around the mean, while a large SD means more variability. SD is the basis for setting Levey-Jennings control limits, such as the mean plus or minus 2 SD.

#94 · Laboratory Mathematics Statistics and Quality Management

Calculate the coefficient of variation (CV%) if the mean of a control is 100 and the SD is 2.

Answer

$CV\% = (SD \div \text{mean}) \times 100$. That is $(2 \div 100) \times 100 = 2\%$. A lower CV% means better precision, so a lab compares its CV% for a control to the manufacturer's expected range to judge how consistent its testing is.

#95 · Laboratory Mathematics Statistics and Quality Management

What is measurement uncertainty?

Answer

Measurement uncertainty is a numeric estimate of the range within which the true value of a result likely falls, given the normal variability of the test method. It reflects that every measurement has some built-in imprecision, even when performed correctly. Labs report or hold this data on file to support clinical interpretation of borderline results.

#96 · Laboratory Mathematics Statistics and Quality Management

Define accuracy in laboratory testing.

Answer

Accuracy describes how close a measured result is to the true or accepted value. A method can be accurate on average even if individual repeated results vary somewhat, as long as they centre on the correct value. Accuracy is usually checked using reference materials with a known concentration.

#97 · Laboratory Mathematics Statistics and Quality Management

Define precision in laboratory testing, and how it differs from accuracy.

Answer

Precision describes how close repeated measurements of the same sample are to each other, regardless of whether they are close to the true value. A method can be precise but inaccurate if it consistently gives the same wrong answer. Accuracy and precision are both needed for a result to be trustworthy.

#98 · Laboratory Mathematics Statistics and Quality Management

What is a detection limit?

Answer

The detection limit is the lowest concentration of an analyte that an assay can reliably distinguish from zero or background noise. Below this limit, the instrument cannot confirm the analyte is truly present. It is different from a reference range, which describes normal healthy values, not the assay's technical floor.

#99 · Laboratory Mathematics Statistics and Quality Management

How does a reference range differ from a critical value?

Answer

A reference range is the expected span of results seen in a healthy population for a given test. A critical value is a result so abnormally high or low that it represents an immediate risk to the patient and must be reported to the care team right away. A result can fall outside the reference range without being critical.

#100 · Laboratory Mathematics Statistics and Quality Management

Give an example that shows the difference between a reference range, a critical value, and a detection limit.

Answer

A serum potassium reference range might be about 3.5 to 5.0 mmol/L, describing normal healthy results. A potassium of 6.8 mmol/L would be a critical value requiring an immediate call to the physician. The assay's detection limit is the lowest potassium concentration the instrument can measure at all, which is a separate technical specification.

#101 · Laboratory Mathematics Statistics and Quality Management

What is the difference between a standard and a control in the laboratory?

Answer

A standard is a solution of precisely known concentration used to calibrate an instrument, establishing the relationship between signal and concentration. A control is a sample of known expected value that is run like a patient sample to verify the instrument and method are working correctly. Standards calibrate, while controls monitor ongoing performance.

#102 · Laboratory Mathematics Statistics and Quality Management

Why can't a control also be used as a calibration standard?

Answer

A control is designed to be tested using the existing calibration, acting as an independent check on whether that calibration is still valid. If the same material were used for both purposes, an error in the material would not be caught because the system would simply calibrate around it. Keeping them separate protects the integrity of the quality checks.

What is a commercial control?

Answer

A commercial control is a quality control material purchased from an outside manufacturer, with an assigned target value and acceptable range for a specific test method. It is manufactured under strict conditions for stability and lot-to-lot consistency. Labs run it regularly to confirm testing accuracy against an independent, standardized source.

What is an in-house pool used for quality control?

Answer

An in-house pool is a quality control material made internally by combining leftover patient specimens with similar characteristics into one larger, homogeneous batch. It is a lower-cost alternative to commercial controls but has no externally certified target value. The lab must establish its own mean and acceptable range through repeated testing.

#105 · Laboratory Mathematics Statistics and Quality Management

What is a blind duplicate patient sample, and why is it useful for quality checks?

Answer

A blind duplicate is a second aliquot of an actual patient sample that is relabeled and resubmitted for testing without staff knowing it is a repeat. Because staff treat it as a brand new specimen, it tests the whole process realistically, not just the instrument. Comparing the two results checks precision under normal working conditions.

#106 · Laboratory Mathematics Statistics and Quality Management

List the three broad categories of potential sources of error in the lab.

Answer

Errors are generally grouped as pre-analytical (before testing, such as wrong tube or poor collection), analytical (during testing, such as instrument or reagent problems), and post-analytical (after testing, such as transcription or reporting mistakes). Most laboratory errors actually occur in the pre-analytical phase. Recognizing which stage an error came from helps target the correct fix.

#107 · Laboratory Mathematics Statistics and Quality Management

Give two examples of pre-analytical sources of error.

Answer

Pre-analytical errors include patient misidentification, improper fasting, hemolysis from a difficult draw, using the wrong tube additive, or delayed transport of a specimen. These occur before the sample ever reaches the analyzer. Because they happen outside the testing instrument, they cannot be detected by internal quality control checks.

#108 · Laboratory Mathematics Statistics and Quality Management

Give two examples of analytical sources of error.

Answer

Analytical errors include expired or improperly stored reagents, an uncalibrated instrument, pipetting inaccuracies, or interference from substances like lipemia or icterus in the sample. These occur during the actual testing process. Regular quality control and maintenance schedules are designed to catch this category of error.

#109 · Laboratory Mathematics Statistics and Quality Management

Solve this ratio and proportion problem: if 3 mL of reagent treats 5 samples, how many mL are needed for 40 samples?

Answer

Set up the proportion $3/5 = x/40$. Cross multiply: $5x = 3 \times 40 = 120$, so $x = 120 \div 5 = 24$ mL. Ratio and proportion problems like this are common when scaling reagent volumes up or down for batch testing.

#110 · Laboratory Mathematics Statistics and Quality Management

A stock reagent is diluted at a ratio of 2 parts stock to 3 parts diluent. What fraction of the final mixture is stock reagent?

Answer

Total parts = $2 + 3 = 5$, so the stock reagent makes up $2/5$, or 40%, of the final mixture. Understanding ratios as parts of a whole, not just a comparison between two numbers, is essential for reading dilution instructions correctly.

Why are exponents (scientific notation) used to report laboratory results?

Answer

Scientific notation lets very large or very small numbers, such as a white blood cell count of 7.5×10^9 /L or a hormone level of 2.3×10^{-10} mol/L, be written compactly and read accurately without counting long strings of zeros. It reduces the chance of a transcription error from a misplaced decimal. Understanding exponent rules also helps when converting between unit prefixes like milli, micro, and nano.

Simplify: what is 4.0×10^3 expressed as a standard number?

Answer

4.0×10^3 means 4.0 multiplied by 1000, which equals 4000. Moving the decimal point three places to the right for a positive exponent is the quick mental shortcut. The reverse move, three places left, applies for a negative exponent like 10^{-3} .

#113 · Laboratory Mathematics Statistics and Quality Management

How many significant figures are in the value 0.00450?

Answer

There are three significant figures: 4, 5, and the trailing zero, because leading zeros before the first nonzero digit are never significant, but a trailing zero after a decimal point is. Reporting the correct number of significant figures shows the true precision of the measurement. Adding extra digits implies more precision than the instrument actually achieved.

#114 · Laboratory Mathematics Statistics and Quality Management

Round 12.346 to three significant figures.

Answer

Look at the fourth significant digit, 4, which is less than 5, so the third digit stays the same: 12.346 rounds to 12.3. If that fourth digit had been 5 or higher, the third digit would round up instead. Consistent rounding rules prevent small formatting differences from becoming reporting discrepancies.

#115 · Laboratory Mathematics Statistics and Quality Management

What does molarity (mol/L) describe about a solution?

Answer

Molarity is the number of moles of solute dissolved per litre of final solution. It lets chemists compare solution strength based on the actual number of molecules present, not just mass. Molarity is calculated as moles of solute divided by litres of solution.

#116 · Laboratory Mathematics Statistics and Quality Management

Calculate the mass of NaCl needed to make 1 L of a 0.9% w/v isotonic saline solution.

Answer

A 0.9% w/v solution means 0.9 g of NaCl per 100 mL, so for 1000 mL (1 L) multiply by 10: $0.9 \text{ g} \times 10 = 9 \text{ g}$ of NaCl. This 0.9% saline is the standard isotonic solution matching the salt concentration of human blood plasma.

#117 · Laboratory Mathematics Statistics and Quality Management

What does 'isotonic' mean when describing a solution used in the lab or clinically?

Answer

An isotonic solution has the same solute concentration, and therefore the same osmotic pressure, as the fluid inside human cells. Because there is no net water movement across the cell membrane, red blood cells placed in an isotonic solution neither swell nor shrink. Normal saline (0.9% NaCl) is the classic isotonic example.

#118 · Laboratory Mathematics Statistics and Quality Management

Explain the difference between a w/v and a v/v percentage solution.

Answer

A weight/volume (w/v) percentage is grams of a solid solute per 100 mL of final solution, such as 10 g of glucose per 100 mL. A volume/volume (v/v) percentage is mL of a liquid solute per 100 mL of final solution, such as 70 mL of alcohol per 100 mL. The unit of the solute (solid vs liquid) determines which formula applies.

#119 · Laboratory Mathematics Statistics and Quality Management

How much solute is needed to prepare 500 mL of a 10% w/v solution?

Answer

A 10% w/v solution contains 10 g per 100 mL, and 500 mL is 5 times that volume, so multiply $10 \text{ g} \times 5 = 50 \text{ g}$ of solute. This is then dissolved and brought up to a final total volume of 500 mL with solvent, not added to 500 mL of solvent.

#120 · Laboratory Mathematics Statistics and Quality Management

What does a w/w percentage solution measure, and when is it typically used?

Answer

A weight/weight (w/w) percentage is grams of solute per 100 g of total solution, used when both the solute and the final solution are measured by mass rather than volume, such as concentrated acids sold by manufacturers. It accounts for solutions where volume can change with temperature but mass does not. Density must be known to convert a w/w value into a volume-based measurement.

#121 · Laboratory Mathematics Statistics and Quality Management

Using $C_1V_1 = C_2V_2$, how much diluent must be added to 10 mL of a 5 mol/L stock to make it 2 mol/L?

Answer

Solve for the final volume: $V_2 = (C_1 \times V_1) \div C_2 = (5 \times 10) \div 2 = 25$ mL total final volume. Since the stock already contributes 10 mL, the diluent added is $25 - 10 = 15$ mL.

#122 · Laboratory Mathematics Statistics and Quality Management

What does a 1:5 dilution mean, and what fraction of the original concentration remains?

Answer

A 1:5 dilution means 1 part sample combined with 4 parts diluent, for a total of 5 parts, so the sample is present at $1/5$, or 20%, of its original concentration. The number after the colon in a ratio-style dilution refers to the total parts, not the parts of diluent alone. Misreading this convention is a common source of dilution error.

#123 · Laboratory Mathematics Statistics and Quality Management

How would you prepare 5 mL of a 1:10 dilution?

Answer

A 1:10 dilution means 1 part sample in 10 total parts, so for a 5 mL final volume, the sample portion is $5 \div 10 = 0.5$ mL. The remaining 4.5 mL is diluent, making 0.5 mL sample plus 4.5 mL diluent equal 5 mL total.

#124 · Laboratory Mathematics Statistics and Quality Management

In a serial 1:2 dilution series starting at 1:2, what are the next three dilutions?

Answer

Each step in a 1:2 serial dilution halves the concentration of the previous tube, so after 1:2 the series continues 1:4, then 1:8, then 1:16. Serial dilutions are used in titers and standard curves where a wide range of concentrations is needed from a small starting volume.

#125 · Laboratory Mathematics Statistics and Quality Management

A sample is diluted 1:20 and the analyzer reads a glucose value of 4.0 mmol/L on the diluted sample. What is the true concentration in the original sample?

Answer

Multiply the diluted result by the dilution factor: $4.0 \text{ mmol/L} \times 20 = 80 \text{ mmol/L}$. Forgetting to multiply back by the dilution factor is a classic transcription error that produces a falsely low reported result.

#126 · Laboratory Mathematics Statistics and Quality Management

What are the common grades of laboratory chemicals, from lowest to highest purity?

Answer

Technical or commercial grade is the least pure and used for cleaning or non-critical tasks, USP/NF grade meets pharmaceutical standards, reagent (ACS) grade is pure enough for most routine lab testing, and analytical or HPLC grade is the highest purity used for sensitive quantitative work. Using a lower grade than a procedure requires can introduce impurities that skew results. The reagent grade needed is specified in the test's standard operating procedure.

#127 · Laboratory Mathematics Statistics and Quality Management

Why does it matter which grade of chemical is used for a given laboratory test?

Answer

Higher purity grades cost more but contain fewer contaminating substances that could interfere with a sensitive assay or damage delicate instrumentation. Using an unnecessarily high grade wastes money, while using too low a grade risks inaccurate results or equipment damage. The procedure manual specifies the minimum acceptable grade for each test.

#128 · Laboratory Mathematics Statistics and Quality Management

What is the difference between distilled water and deionized water?

Answer

Distilled water is purified by boiling water into steam and condensing it, which removes minerals, most organics, and microorganisms. Deionized water is purified by passing water through ion-exchange resins that remove charged mineral ions but do not necessarily remove organic contaminants or microorganisms. The two methods target different classes of contaminants, so labs sometimes combine them for the highest purity water.

#129 · Laboratory Mathematics Statistics and Quality Management

What are laboratory water grades typically called, and how do they differ in purity?

Answer

Laboratory water is commonly classified as Type I (highest purity, used for sensitive analytical work and molecular testing), Type II (general laboratory reagent use), and Type III (less critical uses like glassware rinsing). Each type has defined limits for conductivity, organic content, and microbial growth. Using water of insufficient purity for a sensitive test can introduce interference and false results.

#130 · Laboratory Mathematics Statistics and Quality Management

Why would a laboratory need different grades of water for different tasks?

Answer

Highly sensitive tests such as molecular assays or trace element analysis require ultra-pure Type I water because even trace contaminants can interfere with the result. Less sensitive tasks, like rinsing general glassware, can safely use a lower purity grade, saving cost and production time. Matching water grade to task need is part of good laboratory practice.

#131 · Laboratory Mathematics Statistics and Quality Management

What is a Quality Management System (QMS) in the laboratory?

Answer

A QMS is the organized set of policies, processes, and procedures a laboratory follows to consistently produce accurate, reliable results and to continually improve its performance. It covers everything from specimen collection through result reporting, not just the testing step itself. A strong QMS is required for laboratory accreditation.

#132 · Laboratory Mathematics Statistics and Quality Management

What is the overall goal of applying QMS theory in a laboratory setting?

Answer

The goal is to prevent errors before they happen, catch problems early when they do occur, and drive ongoing improvement rather than simply reacting to mistakes after a patient is affected. It shifts the culture from blaming individuals to fixing systems and processes. This proactive approach protects both patient safety and result reliability.

#133 · Laboratory Mathematics Statistics and Quality Management

What are Quality System Essentials (QSE), and roughly what do they cover?

Answer

QSEs are the core building-block components that together make up a laboratory's Quality Management System, covering areas such as organization, personnel, equipment, process control, documents and records, information management, occurrence management, and facilities and safety. Each essential is assessed separately during accreditation. Together they ensure no part of the testing process is left unmanaged.

#134 · Laboratory Mathematics Statistics and Quality Management

Why does personnel competency count as a Quality System Essential?

Answer

Even the best instruments and procedures fail if the people running them are not properly trained, assessed, and kept current. Personnel management as a QSE covers hiring qualifications, orientation, ongoing competency assessment, and continuing education. Documenting staff competency protects both quality and accreditation compliance.

#135 · Laboratory Mathematics Statistics and Quality Management

What is risk management, and what role does it play in a QMS?

Answer

Risk management is the systematic process of identifying what could go wrong in a laboratory process, judging how likely and how serious each risk is, and putting controls in place to reduce it before harm occurs. Within a QMS, it shifts the focus from reacting to errors toward proactively preventing them. High-risk steps, like specimen mislabeling, typically get the strongest safeguards.

#136 · Laboratory Mathematics Statistics and Quality Management

What is a 'no-blame culture' in laboratory quality management?

Answer

A no-blame culture encourages staff to report errors, near misses, and unsafe conditions honestly without fear of punishment, because the goal is fixing the underlying system, not punishing the individual. This openness allows a laboratory to actually learn about problems instead of having them hidden. It is considered essential for an effective occurrence management program.

#137 · Laboratory Mathematics Statistics and Quality Management

Differentiate between a nonconformance, an adverse event, and a near miss.

Answer

A nonconformance is any deviation from a written procedure or requirement, whether or not it caused harm. An adverse event (or occurrence) is an incident that actually did cause harm or a negative outcome, such as a delayed critical result reaching the physician. A near miss is an error that was caught before it reached the patient or affected the result, offering a warning sign without actual harm.

#138 · Laboratory Mathematics Statistics and Quality Management

What are quality indicators, and why does a laboratory track them?

Answer

Quality indicators are measurable data points, such as specimen rejection rate or turnaround time, that a laboratory monitors over time to judge how well its processes are performing. Tracking trends in these indicators helps detect a slowly worsening problem before it becomes a major failure. They also provide objective evidence of performance during accreditation review.

#139 · Laboratory Mathematics Statistics and Quality Management

What is the difference between a corrective action and a preventive action?

Answer

A corrective action addresses an error that has already happened, fixing the immediate problem and its root cause so it does not recur. A preventive action addresses a potential problem that has not happened yet but was identified as a risk, putting safeguards in place ahead of time. Both are documented as part of the laboratory's continuous improvement process.

#140 · Laboratory Mathematics Statistics and Quality Management

How does the MLA/T support the Quality Management System in daily practice?

Answer

The MLA/T supports the QMS by following written procedures exactly, running and documenting quality control, reporting errors and near misses honestly, and participating in activities like proficiency testing, internal audits, and procedure reviews. Even though this role usually cannot be assessed by simulation alone, it depends on daily habits like accurate documentation and speaking up about problems. Consistent day-to-day compliance by every staff member is what keeps the whole QMS functioning.

#141 · Specimen Procurement Processing and Data Collection

Who can generate a specimen collection request?

Answer

Only an authorized health care professional, such as a physician, nurse practitioner, or other regulated provider with ordering privileges. The MLA/T must never collect a specimen based on a request from an unauthorized source. This protects both patient safety and the legal validity of the test order.

#142 · Specimen Procurement Processing and Data Collection

What should be verified when reviewing a requisition before collection?

Answer

The MLA/T checks that patient demographic information matches the patient, confirms the tests ordered are complete and legible, and reviews the specimen requirements for each test. This step catches errors before blood is drawn, preventing wasted draws and mislabeled samples.

#143 · Specimen Procurement Processing and Data Collection

What is informed consent in the context of specimen collection?

Answer

Informed consent means the patient has been told what procedure will happen, why it is needed, and any risks involved, and they agree to proceed. The MLA/T must obtain this consent before collecting any specimen. A patient always has the right to refuse.

#144 · Specimen Procurement Processing and Data Collection

Why is proper specimen collection important for patient care?

Answer

A poorly collected specimen can produce inaccurate results, leading to wrong diagnoses or treatment decisions. It can also mean the patient must be stuck again, causing unnecessary discomfort and delay. Correct technique protects both patient safety and the reliability of the whole testing process.

#145 · Specimen Procurement Processing and Data Collection

Why does patient and specimen identification matter at every stage of handling?

Answer

A mix-up at collection, labeling, transport, or reporting can attach one patient's result to another patient's chart, which can cause serious harm. Identification must be checked and reconfirmed from the moment the specimen is collected until the result is released. Every handoff is a new chance for error, so every handoff needs a check.

#146 · Specimen Procurement Processing and Data Collection

What are the two unique identifiers used to verify patient identity?

Answer

Common examples are full name and date of birth, or full name and a unique medical record/health card number. Room number or bed location is never acceptable as an identifier because patients can be moved. Both identifiers must be actively confirmed with the patient or their chart, not assumed.

#147 · Specimen Procurement Processing and Data Collection

How should an MLA/T verify identity for a patient who cannot speak or is unconscious?

Answer

The MLA/T uses the identification band on the patient's wrist and cross-checks it against the requisition, since verbal confirmation is not possible. A second staff member or family member can assist in confirming identity if needed. The two-identifier rule still applies using the band information.

#148 · Specimen Procurement Processing and Data Collection

Name common venous collection sites in an adult.

Answer

The median cubital, cephalic, and basilic veins in the antecubital area of the arm are the primary sites. The median cubital vein is preferred because it is generally the largest, most stable, and least painful to puncture. The basilic vein is used last because it lies close to the brachial artery and nerve.

#149 · Specimen Procurement Processing and Data Collection

What is a common capillary collection site for an adult versus an infant?

Answer

In adults, the lateral or medial surface of the fingertip (usually the middle or ring finger) is used. In infants, the medial or lateral plantar surface of the heel is used instead, because a fingerstick risks bone injury in a small foot or hand. Heel sticks avoid the central area of the heel to protect the calcaneus bone.

#150 · Specimen Procurement Processing and Data Collection

What equipment makes up a standard vacutainer venipuncture setup?

Answer

A vacutainer holder, a double-ended needle, and the correctly ordered evacuated collection tubes. Needle gauge is chosen based on vein size and patient factors, commonly 21 or 22 gauge for adults. A tourniquet, alcohol swab, gauze, and bandage complete the standard setup.

#151 · Specimen Procurement Processing and Data Collection

When is a butterfly (winged infusion) needle preferred over a straight needle?

Answer

A butterfly needle is preferred for small, fragile, or difficult veins, such as in pediatric, geriatric, or oncology patients. Its shorter, thinner needle and flexible tubing allow better control during the draw. It is also commonly used for hand veins.

#152 · Specimen Procurement Processing and Data Collection

What is a lancet used for and what depth control does it have?

Answer

A lancet is a small, spring-loaded device used to puncture the skin for capillary collection. Lancets are designed with a fixed puncture depth appropriate to the site, shallower for infant heel sticks and deeper for adult fingersticks. Using the wrong depth device risks either an inadequate sample or unnecessary tissue injury.

#153 · Specimen Procurement Processing and Data Collection

What is the purpose of an anticoagulant additive in a collection tube?

Answer

An anticoagulant prevents the blood from clotting so that whole blood or plasma can be tested, which is required for tests like a CBC or coagulation studies. Different anticoagulants work by different mechanisms, such as chelating calcium or inhibiting thrombin. Using the wrong anticoagulant, or none at all, can invalidate the test result.

#154 · Specimen Procurement Processing and Data Collection

What anticoagulant is used in a lavender-top tube and what is it used for?

Answer

Lavender-top tubes contain EDTA, which chelates calcium to prevent clotting while preserving cell morphology. It is the standard tube for a complete blood count (CBC) and blood film. EDTA is not suitable for coagulation or most chemistry testing.

#155 · Specimen Procurement Processing and Data Collection

What anticoagulant is used in a light blue-top tube and why is the fill volume critical?

Answer

Light blue-top tubes contain sodium citrate and are used for coagulation testing such as PT/INR and PTT. The tube must be filled to the exact draw volume because the citrate-to-blood ratio must stay correct for accurate results. An underfilled citrate tube is a common cause of specimen rejection.

#156 · Specimen Procurement Processing and Data Collection

What happens if the wrong anticoagulant is used for a test?

Answer

The additive can interfere chemically with the analyte being measured, causing falsely high or low results. It can also alter cell shape or clotting activity, making the sample unsuitable for the intended analysis entirely. This is why matching the correct tube to the correct test is a critical safety step, not just a labeling detail.

#157 · Specimen Procurement Processing and Data Collection

What is the standard order of draw for venipuncture, from first to last?

Answer

Blood culture bottles first, then light blue (citrate), then plain red (no additive) or gold/tiger SST (clot activator and gel), then green (heparin), then lavender (EDTA), then gray (fluoride/oxalate). This CLSI-recommended order prevents additive carryover from one tube contaminating the next tube drawn.

#158 · Specimen Procurement Processing and Data Collection

Why does the order of draw put the citrate (coagulation) tube before the EDTA tube?

Answer

If EDTA is drawn before citrate, trace EDTA carryover into the citrate tube can falsely prolong clotting times because EDTA also binds calcium. Drawing citrate early, right after any blood culture, avoids this contamination. Even small amounts of carryover can significantly skew coagulation results.

#159 · Specimen Procurement Processing and Data Collection

What can happen if the order of draw is not followed correctly?

Answer

Additive carryover between tubes can cause falsely elevated or decreased results, especially for electrolytes and coagulation tests. This can lead to an incorrect clinical decision based on a corrupted result. The correct order minimizes cross-contamination between different tube additives.

#160 · Specimen Procurement Processing and Data Collection

List the general steps of a routine venipuncture procedure.

Answer

Verify the order and identify the patient with two identifiers, assemble supplies, apply the tourniquet, select and cleanse the site, perform the puncture, fill tubes in the correct order while releasing the tourniquet, apply pressure, label tubes at the bedside, and dispose of sharps safely. Each step protects either patient safety or specimen quality.

#161 · Specimen Procurement Processing and Data Collection

How long should a tourniquet be left on the arm before release, and why does this matter?

Answer

A tourniquet should generally be released within one minute of application. Leaving it on too long causes hemoconcentration, where fluid shifts out of the vessel and falsely raises concentrations of cells and large molecules. This can distort results for tests like hematocrit and protein-based analytes.

#162 · Specimen Procurement Processing and Data Collection

How should capillary blood collection differ in technique from venous collection?

Answer

Capillary collection uses a lancet puncture rather than a needle, and the first drop of blood is usually wiped away to avoid tissue fluid contamination. Free-flowing drops are collected without excessive squeezing, called milking, because milking dilutes the sample with tissue fluid. Order of draw for capillary tubes differs slightly, generally starting with EDTA-based tests before other additives.

#163 · Specimen Procurement Processing and Data Collection

How should venipuncture be adapted for a patient with difficult venous access?

Answer

The MLA/T should try warming the site to improve blood flow, use a blood pressure cuff instead of a tight tourniquet if needed, and consider a smaller gauge needle or butterfly set. If two attempts fail, protocol usually requires notifying a more experienced staff member rather than continuing to attempt the draw. Patient comfort and safety take priority over completing the draw on the first try.

#164 · Specimen Procurement Processing and Data Collection

Why should blood not be drawn from the arm on the same side as a mastectomy with lymph node removal?

Answer

Lymph node removal can impair lymphatic drainage on that side, increasing the risk of infection, swelling, or lymphedema from a needle puncture. The MLA/T should draw from the unaffected arm whenever possible. If both arms are affected, the ordering provider should be consulted before proceeding.

#165 · Specimen Procurement Processing and Data Collection

Name two potential hazards to the patient during specimen collection.

Answer

Hazards include hematoma formation from improper needle withdrawal or pressure, and nerve injury from incorrect needle angle or site selection. Fainting (syncope) is also a common risk, especially in anxious or first-time patients. The MLA/T should always ask patients to sit or lie down during collection to reduce fall risk.

#166 · Specimen Procurement Processing and Data Collection

Name a hazard to the MLA/T during specimen collection and how it is minimized.

Answer

Accidental needlestick injury is a major hazard, carrying risk of bloodborne pathogen exposure. It is minimized by using safety-engineered needles, never recapping needles by hand, and immediately disposing of sharps into a puncture-resistant sharps container. Standard precautions and proper PPE also reduce splash and contact exposure risk.

#167 · Specimen Procurement Processing and Data Collection

What is the correct after-care once a venipuncture is complete?

Answer

Apply firm, direct pressure to the site with gauze until bleeding stops, then apply a bandage if appropriate. The patient should be observed briefly for signs of dizziness or a developing hematoma before being discharged from the collection area. The arm should not be bent with a cotton ball under pressure, as this does not adequately compress the vessel.

#168 · Specimen Procurement Processing and Data Collection

What should the MLA/T do if a hematoma begins to form during or after a draw?

Answer

Release the tourniquet and remove the needle immediately if it is still forming during the draw, then apply firm pressure for several minutes. Ice can be applied afterward to reduce swelling. The incident should be documented according to organizational procedure.

#169 · Specimen Procurement Processing and Data Collection

What is the correct aseptic technique for blood culture collection?

Answer

The skin site must be disinfected with a stronger antiseptic, such as chlorhexidine, using a scrub-and-dry technique rather than a simple alcohol wipe. The bottle tops are also disinfected before inoculation. This strict technique minimizes the chance of skin flora contaminating the culture and causing a false-positive result.

#170 · Specimen Procurement Processing and Data Collection

In what order are blood culture bottles inoculated relative to other tubes, and between the aerobic and anaerobic bottles themselves?

Answer

Blood culture bottles are always drawn first in the overall order of draw, before any other additive tubes. Between the two bottles, the aerobic bottle is typically filled first to avoid introducing any air bubble from the needle or tubing into the anaerobic bottle. Correct fill volume in each bottle is also critical for reliable growth detection.

#171 · Specimen Procurement Processing and Data Collection

What special documentation is needed for a timed specimen, such as a 2-hour postprandial glucose?

Answer

The exact time of collection must be recorded on both the tube label and the requisition, along with the time of the reference event, such as the meal or medication dose. Collecting even a few minutes off schedule can invalidate the result. The MLA/T should confirm timing with the patient or chart before proceeding.

#172 · Specimen Procurement Processing and Data Collection

What special precaution applies to collecting a legal blood alcohol specimen?

Answer

A non-alcohol-based antiseptic, such as aqueous benzalkonium chloride or povidone-iodine, must be used to clean the site instead of an alcohol swab. Alcohol on the skin could contaminate the sample and be challenged in court. Strict chain-of-custody documentation is also required alongside the collection.

#173 · Specimen Procurement Processing and Data Collection

What is chain of custody in legal specimen testing?

Answer

Chain of custody is a documented, unbroken record of every person who collected, handled, transported, or stored a specimen from collection to final result. It proves the specimen was not tampered with and can be used as evidence. Any gap in the documented chain can make the result inadmissible in legal proceedings.

#174 · Specimen Procurement Processing and Data Collection

What steps are required specifically for legal (forensic) specimen collection beyond routine collection?

Answer

The patient's identity is verified with government-issued photo identification when possible, and the specimen container is sealed with tamper-evident tape immediately after collection. Every transfer of custody is signed and dated on a chain-of-custody form. The collector must also witness that the sample came directly from the named patient.

#175 · Specimen Procurement Processing and Data Collection

What should the MLA/T do when a repeat collection is required due to a rejected specimen?

Answer

The MLA/T follows organizational procedure to notify the appropriate area, confirms the original order is still valid, and recollects using correct technique to avoid repeating the same error. The reason for rejection should be documented so the same mistake is not repeated. The patient should be informed politely why a second collection is needed.

#176 · Specimen Procurement Processing and Data Collection

What is the difference between isolation precautions and reverse isolation?

Answer

Standard isolation precautions protect staff and other patients from an infectious patient by using barriers like gowns, gloves, and masks. Reverse isolation protects a vulnerable, immunocompromised patient from outside pathogens carried by staff or visitors. In reverse isolation, the PPE mainly protects the patient rather than the caregiver.

#177 · Specimen Procurement Processing and Data Collection

What PPE is used for droplet precautions versus airborne precautions?

Answer

Droplet precautions require a standard surgical mask, gown, and gloves for close patient contact, since droplets travel only short distances. Airborne precautions require a fitted N95 respirator, plus gown and gloves, because pathogens can remain suspended in air and travel farther. Contact precautions add gown and gloves to prevent transfer from surfaces or direct skin contact.

#178 · Specimen Procurement Processing and Data Collection

How must a biological specimen be packaged for shipment outside the collecting facility?

Answer

It must use leak-proof primary containment, absorbent material to contain any leakage, and a rigid outer container, following applicable transport of dangerous goods legislation. Biohazard labeling is required on the outer packaging. Temperature control materials, like ice packs, are added if the specimen requires refrigeration during transit.

#179 · Specimen Procurement Processing and Data Collection

What are the collection instructions for a 24-hour urine specimen?

Answer

The patient discards the first morning void and notes that start time, then collects every void for the next 24 hours into a designated container, ending with the first void the next morning at the same start time. All urine passed during the period must be included, and none can be discarded. Some tests require the container to be kept refrigerated or to contain a chemical preservative.

#180 · Specimen Procurement Processing and Data Collection

What is a mid-stream urine (MSU) collection and why is the technique important?

Answer

The patient cleans the genital area, begins urinating into the toilet, then collects a sample partway through the stream into a sterile container without stopping the flow. This technique avoids contamination from skin flora and the initial urethral flush. Poor technique is a common cause of false-positive urine cultures.

#181 · Specimen Procurement Processing and Data Collection

What is required for a fecal specimen collected for ova and parasites (O&P;) testing?

Answer

The sample is collected without contamination from urine or water and is typically placed into preservative vials provided by the lab, since parasites can degrade quickly. Multiple samples on different days may be requested to increase detection sensitivity. No antidiarrheal medications or barium studies should have been used recently, as they can interfere with results.

#182 · Specimen Procurement Processing and Data Collection

How does fecal collection for culture and sensitivity (C&S;) differ from fecal fat collection?

Answer

C&S; collection needs only a small fresh sample in a clean or sterile container to identify a causative bacterial pathogen. Fecal fat testing usually requires a timed collection over 24 to 72 hours into a large container, sometimes with a preceding controlled-fat diet, to quantify fat malabsorption. The two tests have very different volume and timing requirements.

#183 · Specimen Procurement Processing and Data Collection

What are the general requirements for collecting a microbiology specimen?

Answer

The sample must be collected from the actual site of suspected infection using sterile technique to avoid contamination from normal flora. It should be collected before antibiotics are started whenever possible, since antibiotics can suppress bacterial growth. The specimen must be transported promptly in the correct medium to keep organisms viable.

#184 · Specimen Procurement Processing and Data Collection

What is the best technique and timing for a sputum sample submitted for culture and sensitivity?

Answer

The sample should be a deep cough specimen from the lungs, not saliva, ideally collected first thing in the morning before eating or drinking. The patient rinses their mouth with water first to reduce oral contamination. The specimen should be sent to the lab promptly to preserve organism viability.

#185 · Specimen Procurement Processing and Data Collection

How does a sputum sample for cytology differ from one for culture and sensitivity?

Answer

A cytology sputum sample is often collected over consecutive early-morning specimens and is placed into a fixative solution to preserve cell structure for microscopic examination. A culture and sensitivity sample must stay fresh, unfixed, and be processed quickly to allow organisms to grow. Fixative would kill any organisms, making it unsuitable for culture.

#186 · Specimen Procurement Processing and Data Collection

What are the specimen requirements for a fertility (semen analysis) seminal fluid sample?

Answer

The sample is collected by masturbation into a sterile container after a period of sexual abstinence, usually two to five days, and must reach the lab within about one hour at body temperature. Lubricants and condoms are avoided because they can be toxic to sperm. Any spilled portion of the sample should be noted since it affects volume-based results.

#187 · Specimen Procurement Processing and Data Collection

How does seminal fluid collection differ for a post-vasectomy check compared to a fertility study?

Answer

A post-vasectomy sample simply confirms the absence of sperm and is usually collected after a set number of weeks and ejaculations following the surgery, rather than after a period of abstinence. A fertility study requires abstinence beforehand and evaluates sperm count, motility, and morphology. Both require prompt delivery to the lab.

#188 · Specimen Procurement Processing and Data Collection

How should a fine needle aspirate biopsy smear be preserved after collection?

Answer

The aspirated material is smeared thinly and evenly onto a glass slide and either air-dried or immediately fixed in alcohol, depending on the stain the lab requires. Any leftover fluid may be placed into a preservative solution for cell block preparation. Proper fixation prevents cell distortion that could interfere with diagnosis.

#189 · Specimen Procurement Processing and Data Collection

What general handling requirement applies to body fluid specimens like pleural or peritoneal fluid?

Answer

Body fluids are usually collected into sterile containers, and some tests require an anticoagulant tube to prevent clotting before cell counts can be performed. Samples should be transported promptly, as cells can degrade quickly outside the body. The requisition should specify the exact fluid source since testing and reference ranges differ by fluid type.

#190 · Specimen Procurement Processing and Data Collection

What is included in a skin scraping collection kit provided to a patient?

Answer

A kit typically includes a sterile scalpel or blade, a collection container or slide, and clear written instructions for the patient or clinician performing the scraping. Instructions explain how to sample from the active edge of a lesion, where fungal elements are most concentrated. The MLA/T's role is to provide the correct kit and explain proper use.

#191 · Specimen Procurement Processing and Data Collection

What does it mean to handle specimens according to priority and stability?

Answer

STAT or urgent specimens must be processed and delivered ahead of routine samples because a delay can affect patient care. Stability refers to how quickly an analyte degrades, so time-sensitive specimens, like blood gases or ammonia, must be processed immediately rather than left waiting. Sorting specimens correctly on arrival prevents unnecessary result delays or invalid results.

#192 · Specimen Procurement Processing and Data Collection

Name three common criteria for rejecting a specimen upon receipt.

Answer

Common rejection criteria include a mislabeled or unlabeled tube, insufficient sample volume (quantity not sufficient), a clotted sample in an anticoagulant tube, or hemolysis affecting certain tests. Gross contamination or the wrong tube type for the ordered test are also grounds for rejection. Each facility maintains a specific written rejection criteria list.

#193 · Specimen Procurement Processing and Data Collection

What should the MLA/T do after identifying a specimen that meets rejection criteria?

Answer

The specimen should not be tested, and the appropriate area or ordering provider should be notified following organizational procedure. The reason for rejection is documented, and a recollection is typically requested. Patient safety depends on never reporting a result from a compromised sample.

#194 · Specimen Procurement Processing and Data Collection

What does 'accessioning' a specimen mean?

Answer

Accessioning is the process of officially logging a specimen into the laboratory information system upon receipt, assigning it a unique tracking number linked to the patient and test order. This creates a traceable record from arrival through result reporting. Accessioning also confirms the specimen matches its requisition before further processing begins.

#195 · Specimen Procurement Processing and Data Collection

What is the general procedure for performing serum or plasma separation?

Answer

The tube is centrifuged at the manufacturer-specified speed and time to separate cells from the liquid fraction, then the serum or plasma is carefully removed or the gel barrier is left to maintain separation. Serum comes from clotted tubes, while plasma comes from anticoagulated tubes that were never allowed to clot. Improper centrifugation can cause hemolysis or incomplete separation.

#196 · Specimen Procurement Processing and Data Collection

Why must a serum separator tube clot completely before centrifugation?

Answer

If centrifuged before full clotting, usually about 30 minutes at room temperature, fibrin strands can still form after spinning and interfere with the analyzer probe. This can cause inaccurate results or instrument errors. Following the manufacturer's recommended clotting time prevents this problem.

#197 · Specimen Procurement Processing and Data Collection

What general storage temperature ranges are used for laboratory specimens, and how are they chosen?

Answer

Common ranges are room temperature, refrigerated around 2 to 8 degrees Celsius, and frozen at minus 20 degrees Celsius or lower, depending on how quickly the analyte degrades. The requisition or laboratory procedure manual specifies which storage condition applies to each test. Using the wrong storage temperature can cause an analyte to break down before testing occurs.

#198 · Specimen Procurement Processing and Data Collection

What does processing samples for quality control involve?

Answer

Quality control samples with known expected values are run alongside patient samples to confirm the testing process and equipment are working correctly. Results falling outside the acceptable range indicate a problem that must be resolved before patient results are released. This step is part of ensuring every reported result can be trusted.

#199 · Specimen Procurement Processing and Data Collection

What special handling is often required when processing a sample for basic molecular techniques?

Answer

Molecular specimens often require strict anti-contamination precautions, since even trace amounts of foreign DNA or RNA can produce a false result. Dedicated pipettes, workspaces, and sometimes specific collection tubes are used to protect sample integrity. Samples may also need to be processed or frozen more quickly than routine chemistry specimens to preserve nucleic acid quality.

#200 · Specimen Procurement Processing and Data Collection

What is hemolysis and how can it interfere with test results?

Answer

Hemolysis is the rupture of red blood cells, releasing their intracellular contents into the surrounding serum or plasma. This falsely elevates analytes concentrated inside cells, such as potassium and certain enzymes, and can also interfere optically with some assays. Hemolysis is commonly caused by a traumatic draw, excessive tube shaking, or a needle gauge that is too small.

#201 · Specimen Procurement Processing and Data Collection

If an MLA/T notices a possible interference affecting a sample's quality, what is the correct action?

Answer

The MLA/T should recognize and describe the deficiency, such as lipemia, hemolysis, or icterus, but refer the interpretation and next steps to an MLT rather than deciding independently. This respects scope of practice while still flagging the concern promptly. Timely reporting prevents a compromised result from being released.

#202 · Specimen Procurement Processing and Data Collection

What is tracked by a laboratory information system (LIS) from test request through to reporting?

Answer

An LIS manages the entire specimen journey: the initial test request, requisition details, specimen collection and receipt, linkage to the patient record, and ultimately the electronic medical record (EMR) where results are released. It allows staff at each stage to see the specimen's current status. This continuous tracking supports both efficiency and patient safety.

#203 · Specimen Procurement Processing and Data Collection

What is a critical value and what must the MLA/T do upon recognizing one?

Answer

A critical value is a result so far outside the normal range that it represents an immediate risk to the patient's life if not acted on quickly. The MLA/T is not authorized to interpret or release it, but must recognize it and refer it promptly to an MLT for verification and urgent notification to the care team. Delayed recognition of a critical value can directly harm a patient.

#204 · Specimen Procurement Processing and Data Collection

What should the MLA/T do if someone calls asking for a patient's lab results but is not authorized to receive them?

Answer

The MLA/T must decline to release any patient information, since confidentiality regulations only permit sharing results with authorized individuals such as the ordering provider or the patient through approved channels. The request should be documented and escalated according to organizational policy if needed. Protecting patient confidentiality applies regardless of how the request is phrased or how urgent it sounds.

#205 · Specimen Procurement Processing and Data Collection

What are the general rules for document retention and disposal of laboratory records?

Answer

Laboratory records, including requisitions and result reports, must be kept for a legally and organizationally defined retention period before disposal, which varies by document and jurisdiction. When records are disposed of, it must be done securely, such as shredding, to protect patient confidentiality. Following retention and disposal rules supports both legal compliance and ongoing patient care continuity.

#206 · Laboratory Safety

What does the Occupational Health and Safety Act (OHSA) govern in a medical laboratory?

Answer

OHSA sets the legal minimum standards for worker health and safety in the workplace. It defines the duties of employers, supervisors, and workers, including the worker's right to know about hazards, the right to participate in safety decisions, and the right to refuse unsafe work. Laboratories must follow OHSA alongside any sector-specific safety regulations.

#207 · Laboratory Safety

What is WHMIS 2015?

Answer

WHMIS 2015 is Canada's Workplace Hazardous Materials Information System, aligned with the Globally Harmonized System (GHS) for classifying and communicating chemical hazards. It requires hazardous products to have standardized labels, pictograms, and Safety Data Sheets. It replaced the older WHMIS 1988 system to create consistency with international hazard communication standards.

#208 · Laboratory Safety

What do the Hazardous Products Regulations (HPR) require of suppliers?

Answer

The HPR is the federal regulation that sets the technical rules for classifying hazardous products and preparing WHMIS labels and Safety Data Sheets. Suppliers and manufacturers must classify products correctly and provide compliant labeling and SDS information before selling or importing them into Canada. It works together with WHMIS 2015 to protect workers who handle these products.

#209 · Laboratory Safety

What does the Transportation of Dangerous Goods Act and Regulations (TDG) cover?

Answer

TDG governs the safe handling, packaging, labeling, and shipping of dangerous goods, including laboratory chemicals and biological specimens, when they are transported by road, rail, air, or water. Lab staff who ship or receive regulated substances need TDG training. Rules include proper classification, documentation, and use of approved containers.

#210 · Laboratory Safety

What does the Nuclear Safety and Control Act regulate in a lab setting?

Answer

This act and its associated regulator, the Canadian Nuclear Safety Commission, oversee the safe use, handling, storage, and disposal of radioactive materials in Canada. Laboratories that use radioisotopes for testing must follow licensing requirements and radiation safety protocols under this act. It exists to protect workers, the public, and the environment from radiation risk.

#211 · Laboratory Safety

What is the purpose of the Human Pathogens and Toxins Act (HPTA)?

Answer

The HPTA regulates activities involving human pathogens and toxins to protect the health and safety of the public and laboratory workers. It requires licensing for facilities that handle certain risk group organisms and sets biosafety and biosecurity requirements. Medical laboratories handling infectious specimens must comply with its containment and reporting rules.

#212 · Laboratory Safety

Why is understanding safety legislation part of an MLA/T's competency?

Answer

The MLA/T must be able to work within the legal framework that governs laboratory operations, not just follow local policy. Knowing which act applies to a given hazard, such as chemical, biological, or radioactive, helps the assistant respond correctly and understand why certain procedures exist. This knowledge supports both compliance and worker protection.

#213 · Laboratory Safety

What are routine practices in infection control?

Answer

Routine practices are the basic infection prevention measures used with every patient and specimen, regardless of known infection status, because any sample could be infectious. They include hand hygiene, appropriate PPE use, safe handling of sharps, and proper cleaning of work surfaces. This approach assumes all blood and body fluids are potentially hazardous.

#214 · Laboratory Safety

What should an MLA/T do first when noticing a safety issue in the lab?

Answer

The assistant should stop the activity if there is immediate danger, then report the issue to a supervisor following institutional protocol. Depending on the hazard, this may include isolating the area, using appropriate PPE, or activating an emergency procedure. Prompt reporting helps prevent injury to coworkers and allows timely correction of the hazard.

#215 · Laboratory Safety

What is the difference between routine practices and additional precautions?

Answer

Routine practices are used with every patient at all times as a baseline standard. Additional precautions are extra measures added on top of routine practices for patients known or suspected to have specific transmissible infections, such as contact, droplet, or airborne precautions. The extra measures may include specialized PPE or isolation procedures.

#216 · Laboratory Safety

General safe laboratory practice: why should eating and drinking never occur in the lab work area?

Answer

Eating, drinking, applying cosmetics, or handling contact lenses in the lab risks contaminating hands, food, or mucous membranes with biological or chemical hazards present on surfaces or specimens. These activities must be confined to designated break areas away from testing benches. This is a core routine practice for preventing accidental exposure.

#217 · Laboratory Safety

What is the correct general order for donning PPE?

Answer

A common donning sequence is gown first, then mask or respirator, then eye protection such as goggles or a face shield, and gloves last. Putting gloves on last protects them from contamination while adjusting the other equipment. Following institutional policy for exact order is important since sequences can vary slightly by site.

#218 · Laboratory Safety

What is the correct general order for doffing PPE and why does it matter?

Answer

A common doffing sequence is gloves first, then eye protection, then gown, then mask or respirator last, with hand hygiene performed between steps and after removal. Removing gloves first prevents spreading contamination from the most soiled item to skin or clothing. Careful doffing order is critical because most self-contamination happens during removal, not while wearing PPE.

#219 · Laboratory Safety

Why are gloves removed before other PPE items in most doffing sequences?

Answer

Gloves are typically the most contaminated PPE item because they directly contact specimens and surfaces. Removing them early, using a technique that avoids touching the outer surface with bare skin, reduces the risk of transferring contamination to the face or clothing during removal of the remaining items. Hand hygiene should follow glove removal.

#220 · Laboratory Safety

What PPE is typically required for routine specimen handling in a medical laboratory?

Answer

Standard PPE usually includes a fluid-resistant lab coat or gown, gloves, and eye protection when there is risk of splash. Closed-toe shoes and, in some areas, a face mask may also be required. The exact combination depends on the task and the level of exposure risk involved.

#221 · Laboratory Safety

What is a safety needle and why is it used?

Answer

A safety needle is a sharp device with an engineered safety feature, such as a retractable needle or a hinged shield, that covers or retracts the sharp point after use. It is designed to reduce the risk of accidental needlestick injuries to healthcare workers. Their use is often required by law in Canadian workplaces handling sharps.

#222 · Laboratory Safety

What are the rules for using a sharps container correctly?

Answer

Sharps containers must be puncture-resistant, labeled as biohazardous, and located close to the point of use so sharps are disposed of immediately after use without recapping needles. They should never be filled past the manufacturer's fill line, typically about three-quarters full, and must be closed and replaced once full. Overfilled containers increase the risk of injury during disposal.

#223 · Laboratory Safety

What is an acid bottle carrier used for?

Answer

An acid bottle carrier is a protective handle or basket device used to transport bottles of corrosive acids and other hazardous liquids safely between locations in the lab. It reduces the risk of dropping or spilling the container, which could cause a chemical burn or exposure. It should be used any time a large or heavy chemical bottle is moved.

#224 · Laboratory Safety

What is the purpose of a fume hood in a laboratory?

Answer

A fume hood is a ventilated enclosure that draws hazardous vapors, gases, or fine particles away from the worker and exhausts them safely, protecting the worker from chemical inhalation exposure. It is used when handling volatile or toxic chemicals. The sash should be kept at the recommended height to maintain proper airflow.

#225 · Laboratory Safety

What is the purpose of a biological safety cabinet (BSC), and how does it differ from a fume hood?

Answer

A biological safety cabinet uses HEPA-filtered airflow to protect the worker, the specimen, and sometimes the environment from exposure to infectious aerosols and biological hazards. Unlike a fume hood, which is designed mainly for chemical fumes and does not filter air for return to the room in the same way, a BSC is specifically designed for biosafety work with pathogens.

#226 · Laboratory Safety

What are safety cans used for in the laboratory?

Answer

Safety cans are approved metal or specially designed containers used to store and dispense flammable liquids safely. They typically have a spring-loaded, self-closing lid and a flame arrestor screen to prevent ignition. Using safety cans reduces the risk of fire and vapor buildup when handling flammable solvents.

#227 · Laboratory Safety

What is the purpose of a flammable storage cabinet?

Answer

A flammable storage cabinet is a fire-rated enclosure used to store flammable and combustible liquids away from ignition sources, helping contain a fire if one occurs and slowing heat transfer to the stored chemicals. Cabinets should be properly labeled, kept closed when not in use, and not overloaded beyond their rated capacity.

#228 · Laboratory Safety

Why are gas cylinder carts used, and what safety features do they have?

Answer

Gas cylinder carts are designed to transport compressed gas cylinders safely, keeping them secured upright with a chain or strap to prevent tipping. A tipped cylinder can become a dangerous projectile if the valve breaks off. The cylinder cap should remain on during transport whenever possible.

#229 · Laboratory Safety

What is the purpose of an eyewash station in the lab, and how long should it be used after a chemical splash to the eyes?

Answer

An eyewash station provides a continuous, low-pressure stream of water to flush chemicals, biological material, or debris from the eyes after accidental exposure. Standard practice is to flush the affected eye or eyes for at least 15 minutes while holding the eyelids open. Emergency personnel should be notified during or immediately after flushing.

#230 · Laboratory Safety

When should an emergency shower be used in the laboratory?

Answer

An emergency shower is used when a chemical or hazardous substance contacts a large area of the body or clothing, requiring immediate full-body rinsing to dilute and remove the substance. Contaminated clothing should be removed while flushing continues. The affected person should rinse for the time specified in institutional policy, often at least 15 minutes.

#231 · Laboratory Safety

What does a laboratory spill kit typically contain and when is it used?

Answer

A spill kit usually contains absorbent material, neutralizing agents, PPE, a scoop or dustpan, and a disposal bag, used to safely contain and clean up chemical or biological spills. The kit type must match the hazard, since chemical and biohazard spill kits are not interchangeable. Staff should be trained on the correct spill response before an incident occurs.

#232 · Laboratory Safety

What should a laboratory first aid kit be used for, and what is a key limitation?

Answer

A first aid kit provides supplies for treating minor injuries, such as small cuts or burns, until further medical care is available. It is not a substitute for professional medical evaluation after significant exposures like needlestick injuries or major chemical contact, which require institutional incident procedures and often occupational health follow-up.

#233 · Laboratory Safety

What is the difference between a fire extinguisher and a fire blanket, and when is each used?

Answer

A fire extinguisher is used to put out small fires by smothering, cooling, or interrupting the chemical reaction, and different extinguisher types match different fire classes. A fire blanket is used to smother flames on a person's clothing or to cover small fires by cutting off oxygen. Staff should know the location and correct use of both before an emergency occurs, and remember the PASS technique for extinguishers: Pull the pin, Aim at the base, Squeeze the handle, Sweep side to side.

#234 · Laboratory Safety

When should an MLA/T perform basic CPR, and why is this skill part of laboratory safety competencies?

Answer

Basic CPR is performed on a person who is unresponsive and not breathing normally, such as after a severe reaction, electrical shock, or collapse in the lab, to maintain blood flow to vital organs until emergency responders arrive. MLA/Ts are expected to hold current CPR certification and respond promptly, since laboratory incidents can occasionally involve a coworker or patient in cardiac or respiratory arrest.

#235 · Laboratory Safety

What is Containment Level 1 in laboratory biosafety?

Answer

Containment Level 1 applies to work with lower-risk agents that pose minimal or no hazard to healthy workers or the environment, such as many non-pathogenic organisms. Basic laboratory practices, standard PPE, and routine cleaning are generally sufficient. It represents the baseline biosafety level with no special engineering controls required.

#236 · Laboratory Safety

What is Containment Level 2 in laboratory biosafety, and how does it differ from Level 1?

Answer

Containment Level 2 applies to moderate-risk agents that can cause disease in humans but are generally treatable and have a low risk of spreading in the community, such as many clinical specimens. It requires additional controls beyond Level 1, such as use of a biological safety cabinet for procedures that generate aerosols, restricted access, and enhanced training. Most clinical diagnostic labs operate at this level.

#237 · Laboratory Safety

Why should any patient specimen be treated as a potential biological hazard?

Answer

Because a patient's infection status is often unknown or unconfirmed at the time of testing, routine practices require all blood, body fluids, and specimens to be handled as if they are infectious. This standard approach protects workers consistently rather than relying on assumptions about which samples are actually hazardous.

#238 · Laboratory Safety

What are the nine WHMIS 2015 hazard pictograms?

Answer

The nine WHMIS pictograms are Flame, Flame Over Circle, Exploding Bomb, Corrosion, Gas Cylinder, Skull and Crossbones, Health Hazard, Exclamation Mark, and Biohazardous Infectious Materials. Each pictogram represents a specific category of physical, health, or biological hazard. Products may display more than one pictogram if they have multiple hazards.

#239 · Laboratory Safety

What does the WHMIS Flame pictogram indicate?

Answer

The Flame pictogram indicates a material is flammable, meaning it can easily ignite and burn, such as flammable liquids, gases, aerosols, or solids. It can also indicate self-reactive or pyrophoric substances. Products with this symbol should be stored away from ignition sources, often in a flammable storage cabinet.

#240 · Laboratory Safety

What does the WHMIS Flame Over Circle pictogram indicate?

Answer

The Flame Over Circle pictogram indicates an oxidizing material, meaning it can cause or intensify a fire by providing oxygen, even though the substance itself may not be flammable. Oxidizers should be stored away from flammable and combustible materials. Contact with an oxidizer can dramatically increase fire risk.

#241 · Laboratory Safety

What does the WHMIS Exploding Bomb pictogram indicate?

Answer

The Exploding Bomb pictogram warns that a substance is explosive or self-reactive and may explode under certain conditions such as heat, shock, or friction. These materials require special handling and storage precautions to avoid triggering an unintended reaction. This hazard class is less common in a typical clinical lab but may appear with certain reagents.

#242 · Laboratory Safety

What does the WHMIS Corrosion pictogram indicate?

Answer

The Corrosion pictogram indicates a substance can cause severe skin burns, eye damage, or can corrode metals, such as strong acids and bases. Corrosive chemicals should be handled with appropriate PPE, including gloves and eye protection, and transported using devices like an acid bottle carrier.

#243 · Laboratory Safety

What does the WHMIS Gas Cylinder pictogram indicate?

Answer

The Gas Cylinder pictogram indicates the product is a compressed, liquefied, or dissolved gas under pressure, which could explode if heated or damaged. Cylinders bearing this symbol should always be secured upright, transported on a gas cylinder cart, and capped when not in use.

#244 · Laboratory Safety

What does the WHMIS Skull and Crossbones pictogram indicate?

Answer

The Skull and Crossbones pictogram indicates acute toxicity, meaning the substance can cause serious health effects or death after a single or short-term exposure through swallowing, skin contact, or inhalation. These products require strict handling precautions and immediate first aid response if exposure occurs.

#245 · Laboratory Safety

What does the WHMIS Health Hazard pictogram indicate?

Answer

The Health Hazard pictogram indicates chronic and serious health effects such as carcinogenicity, respiratory sensitization, reproductive toxicity, or organ toxicity from repeated or long-term exposure. It differs from the Skull and Crossbones symbol, which covers acute, immediate toxicity rather than long-term effects.

#246 · Laboratory Safety

What does the WHMIS Exclamation Mark pictogram indicate?

Answer

The Exclamation Mark pictogram indicates a less severe health hazard than the Health Hazard or Skull and Crossbones symbols, such as skin or eye irritation, skin sensitization, or drowsiness. It is a caution that the substance can still cause discomfort or mild harm and should be handled carefully.

#247 · Laboratory Safety

What does the WHMIS Biohazardous Infectious Materials pictogram indicate?

Answer

This pictogram indicates a material contains organisms or their toxins known to cause disease in humans or animals, such as certain clinical specimens or cultures. It signals the need for biosafety precautions, including appropriate containment level practices and PPE.

#248 · Laboratory Safety

What are the mandatory sections of a WHMIS-compliant supplier label?

Answer

A supplier label must include the product identifier, initial supplier identifier, hazard pictograms, signal word, hazard statements, precautionary statements, and a statement that a Safety Data Sheet is available. This standardized format ensures workers can quickly identify key hazard information before handling a product.

#249 · Laboratory Safety

What is a Safety Data Sheet (SDS) and what information does it provide?

Answer

An SDS is a standardized 16-section document supplied by the manufacturer that provides detailed information about a hazardous product, including its hazards, composition, safe handling, storage, first aid measures, and spill response. Workers should know where SDSs are kept and how to access them quickly during an emergency.

#250 · Laboratory Safety

What are general rules for safe chemical storage in the laboratory?

Answer

Chemicals should be stored according to compatibility, keeping incompatible substances such as acids and bases or oxidizers and flammables separated. Containers should be labeled clearly, stored below eye level when heavy, and kept in appropriate cabinets such as flammable storage cabinets. Segregation prevents dangerous reactions if a container leaks or breaks.

#251 · Laboratory Safety

What precautions apply to handling compressed gases in the lab?

Answer

Compressed gas cylinders must be secured upright with a chain or strap at all times, transported using a proper cart with the cap on, and stored away from heat sources. Regulators and valves should be checked for leaks, and cylinders should never be dropped or used as rollers, since damage to the valve can turn the cylinder into a projectile.

#252 · Laboratory Safety

What precautions apply to handling cryogenic fluids such as liquid nitrogen?

Answer

Cryogenic fluids are extremely cold and can cause severe frostbite or cold burns on contact with skin, so insulated gloves, a face shield, and closed-toe shoes are required. These fluids should be handled in well-ventilated areas, since rapid vaporization can displace oxygen and create an asphyxiation risk in enclosed spaces.

#253 · Laboratory Safety

What is the correct response if clothing catches fire in the laboratory?

Answer

The person should stop, drop, and roll, or use a fire blanket to smother the flames if one is nearby, rather than running, which can fan the flames. If an emergency shower is close by, using it can also extinguish the fire and cool burned skin. Once flames are out, first aid and medical evaluation should follow immediately.

#254 · Laboratory Safety

What electrical safety precautions should be followed in the laboratory?

Answer

Staff should inspect cords and plugs for damage before use, avoid overloading outlets, keep liquids away from electrical equipment, and never operate equipment with wet hands. Damaged equipment should be removed from service and reported rather than repaired informally. These precautions reduce the risk of shock or electrical fire.

#255 · Laboratory Safety

What is a noise hazard in the laboratory, and how is it managed?

Answer

A noise hazard occurs when equipment, such as centrifuges or analyzers, produces sound levels that can damage hearing with prolonged exposure. Management includes engineering controls like equipment enclosures, limiting exposure time, and providing hearing protection when noise levels exceed safe thresholds. Regular equipment maintenance can also reduce excess noise.

#256 · Laboratory Safety

What is ionizing radiation?

Answer

Ionizing radiation has enough energy to remove electrons from atoms, creating charged particles that can damage cells and DNA. Examples include X-rays and radioactive decay particles such as alpha, beta, and gamma radiation. It poses a greater biological risk than non-ionizing radiation and requires monitoring and shielding precautions.

#257 · Laboratory Safety

What is non-ionizing radiation?

Answer

Non-ionizing radiation has lower energy and is not able to remove electrons from atoms, so it does not directly damage DNA the way ionizing radiation can. Examples include ultraviolet light, visible light, and radiofrequency waves. It can still cause harm, such as UV light causing skin or eye damage, but through different mechanisms than ionizing radiation.

#258 · Laboratory Safety

How is radiation exposure monitored in a laboratory that uses radioactive materials?

Answer

Workers who may be exposed to radiation typically wear a personal dosimeter, such as a badge, that measures cumulative radiation exposure over time. Dosimeter readings are regularly reviewed to ensure exposure stays below regulatory limits. Area monitoring with survey meters may also be used to check for contamination or elevated radiation levels.

#259 · Laboratory Safety

What is the goal of setting up a workstation with proper ergonomics?

Answer

Proper ergonomics aims to minimize the risk of musculoskeletal injury by adjusting chair height, monitor position, keyboard placement, and work surface height to fit the worker's body and task. Good ergonomic setup reduces strain from repetitive motions and prolonged awkward postures common in laboratory work such as pipetting or microscope use.

#260 · Laboratory Safety

What are common ergonomic risk factors for MLA/Ts, and how can they be reduced?

Answer

Common risk factors include repetitive pipetting, prolonged microscope use, awkward wrist positions, and standing or sitting for long periods. These can be reduced by taking regular breaks, using adjustable equipment, keeping frequently used items within easy reach, and maintaining a neutral wrist and back posture during tasks.

#261 · Laboratory Safety

What is the purpose of a steam autoclave in the laboratory?

Answer

A steam autoclave uses pressurized steam at high temperature to sterilize equipment and decontaminate biohazardous waste before disposal, killing bacteria, viruses, and spores. It is commonly used to treat contaminated materials so they can be safely handled as general waste afterward, following institutional validation procedures to confirm sterilization was effective.

#262 · Laboratory Safety

What conditions are typically needed for effective autoclave sterilization?

Answer

Effective sterilization generally requires reaching a set temperature, commonly around 121 degrees Celsius, under pressure for a sufficient exposure time, often 15 to 30 minutes depending on load size and type. Indicators or biological monitors are used to confirm the cycle achieved proper sterilization conditions.

#263 · Laboratory Safety

How should general laboratory waste be handled compared to biomedical waste?

Answer

General waste, such as paper and packaging with no biological or chemical contamination, can go into regular trash streams. Biomedical waste, including specimens, contaminated materials, and sharps, must be segregated into labeled biohazard containers and follow specific institutional and government disposal regulations, since mixing the two creates unnecessary hazard and regulatory violations.

#264 · Laboratory Safety

How should chemical, biological, and radioactive waste be segregated for disposal?

Answer

Each waste type requires its own labeled, compatible container because mixing them can create dangerous reactions or complicate proper treatment. Chemical waste is sorted by compatibility, biological waste goes into biohazard bags or containers, and radioactive waste is stored separately following decay-in-storage or licensed disposal procedures. Following institutional policy for each stream is essential.

#265 · Laboratory Safety

What is biomedical waste and what are its main categories?

Answer

Biomedical waste is waste generated from patient care or laboratory testing that may be contaminated with blood, body fluids, or infectious material. Main categories include sharps such as needles and lancets, human specimens and tissue, and other contaminated materials requiring storage in leak-proof, labeled biohazard containers until proper treatment or disposal.

#266 · Laboratory Safety

What are the two main treatment options for biomedical waste mentioned in MLA/T competencies?

Answer

The two main treatment options are autoclaving, which uses pressurized steam to sterilize waste, and chemical decontamination, which uses disinfecting agents to inactivate pathogens before disposal. The chosen method depends on the type of waste and institutional protocols, and treated waste must meet regulatory standards before leaving the facility.

#267 · Laboratory Safety

What is the correct general response to a biological spill in the laboratory?

Answer

The area should be evacuated of unnecessary personnel, appropriate PPE put on, and the spill kit used to cover the spill with absorbent material and disinfectant, allowing appropriate contact time before cleanup. Contaminated materials are then placed in biohazard waste containers, and the incident is reported following institutional policy.

#268 · Laboratory Safety

What is the correct general response to a chemical spill in the laboratory?

Answer

The worker should alert others, avoid inhaling fumes, and use the appropriate chemical spill kit to neutralize or absorb the spill according to the chemical's SDS instructions, wearing correct PPE throughout. Large or unknown chemical spills may require evacuation and specialized response rather than staff cleanup. The incident must be documented per institutional policy.

#269 · Laboratory Safety

What should be done if a specimen container is found leaking during transport or processing?

Answer

The leaking specimen should be handled with appropriate PPE, placed inside a secondary biohazard bag or container to prevent further contamination, and the outside of the original container and any contaminated surfaces should be disinfected. The incident should be reported and documented, and the specimen's usability for testing should be assessed following lab policy.

#270 · Laboratory Safety

What should an MLA/T do immediately after a needlestick injury, and why is incident reporting required?

Answer

The worker should immediately wash the site with soap and water, encourage bleeding if appropriate, and report to a supervisor and occupational health for risk assessment and possible post-exposure follow-up. Incident reporting is required by policy and legislation because it triggers timely medical evaluation, documents the event for compliance, and helps identify patterns that can prevent future injuries.

#271 · Laboratory Equipment and Supplies

What is the first rule for operating any piece of laboratory equipment?

Answer

Always operate equipment correctly, safely, and according to the manufacturer's or the lab's written procedure. Skipping steps or improvising can produce unsafe conditions and unreliable results.

#272 · Laboratory Equipment and Supplies

Why does every instrument need scheduled basic maintenance, not just repairs when it breaks?

Answer

Basic maintenance catches small problems before they cause equipment failure or inaccurate results. It also keeps the instrument running within the performance limits the manufacturer specifies.

#273 · Laboratory Equipment and Supplies

What must be loaded onto automated and manual instruments along with patient specimens?

Answer

Quality control samples must be loaded alongside patient specimens. This confirms the instrument is working correctly before patient results are reported.

#274 · Laboratory Equipment and Supplies

What is minor troubleshooting on lab equipment, and when should it be done?

Answer

Minor troubleshooting means identifying and correcting simple problems, such as an error message or a jammed sample, using approved steps. It is done whenever equipment is not performing as expected, before continuing to run patient samples.

#275 · Laboratory Equipment and Supplies

How does temperature extreme affect laboratory glassware versus plasticware?

Answer

Glassware generally tolerates high heat well but can crack under sudden temperature changes, while many plastics warp or deform at high temperatures. Choosing the right material for the temperature involved prevents damage and contamination.

#276 · Laboratory Equipment and Supplies

Why can solvents and corrosive chemicals affect the choice between glass and plastic labware?

Answer

Some solvents dissolve or cloud certain plastics, while some corrosive chemicals etch or react with glass. The MLA/T must know the chemical resistance of each material to pick the correct container for the substance being used.

#277 · Laboratory Equipment and Supplies

How should an MLA/T select labware for a task?

Answer

Labware, including glassware and micropipettes, must be appropriate to the specific task at hand. This means matching the material, volume, and tolerance of the item to what the procedure actually requires.

#278 · Laboratory Equipment and Supplies

What does a TD pipette mean, and how is it different from a TC pipette?

Answer

TD stands for to deliver, meaning the pipette is calibrated to dispense the exact stated volume. TC stands for to contain, meaning the pipette holds the stated volume but some liquid remains inside after dispensing and must not be blown out.

#279 · Laboratory Equipment and Supplies

What does the frosted ring on a frosted ring pipette indicate?

Answer

The frosted ring identifies the pipette as a blow-out type, meaning the last small amount of liquid remaining in the tip after draining must be deliberately blown out to deliver the full volume printed on the pipette. Skipping this step would leave the delivered volume short of what the pipette is calibrated for.

#280 · Laboratory Equipment and Supplies

Name the main parts of a compound microscope an MLA/T must identify.

Answer

Key parts include the ocular lens, objective lenses, stage, condenser, iris diaphragm, coarse and fine focus knobs, and light source. Each part has a specific function in forming and adjusting the magnified image.

#281 · Laboratory Equipment and Supplies

What is the function of the objective lenses on a compound microscope?

Answer

The objective lenses sit closest to the specimen and provide the first stage of magnification. Different objectives, such as low, high, and oil immersion power, are switched depending on the detail needed.

#282 · Laboratory Equipment and Supplies

What is the function of the condenser on a compound microscope?

Answer

The condenser focuses and directs light from the light source up through the specimen on the stage. Proper condenser adjustment is essential for a clear, evenly lit image.

#283 · Laboratory Equipment and Supplies

What routine care keeps a compound microscope in good working condition?

Answer

General care includes cleaning the lenses with proper lens paper, covering the microscope when not in use, storing it away from dust and moisture, and never forcing the focus knobs. Regular cleaning prevents scratches and buildup that distort the image.

#284 · Laboratory Equipment and Supplies

What is the purpose of Kohler illumination?

Answer

Kohler illumination is a method of adjusting the microscope's light path so the specimen is evenly and brightly lit with no glare or shadow from the light source itself. It improves image clarity and contrast for accurate observation.

#285 · Laboratory Equipment and Supplies

How is Kohler illumination generally achieved on a compound microscope?

Answer

The field diaphragm and condenser are adjusted so the light source is focused precisely at the level of the condenser, not at the specimen plane, evening out brightness across the field of view. This gives the sharpest, most uniform image for accurate examination.

#286 · Laboratory Equipment and Supplies

What does point of care testing (POCT) mean?

Answer

Point of care testing means performing a lab test, such as blood glucose, right at or near the patient rather than sending the sample to the main lab. It gives a fast result to support immediate clinical decisions.

#287 · Laboratory Equipment and Supplies

What training and certification requirements apply to staff who perform point of care testing?

Answer

Operators must complete required training before performing POCT and must maintain their certification through ongoing competency checks. This ensures results stay reliable even though testing happens outside the main lab.

#288 · Laboratory Equipment and Supplies

What is instrument verification in the context of point of care testing, such as a glucometer?

Answer

Instrument verification means confirming the point of care device is giving accurate results, often by running quality control material or comparing results to a reference method. This must be done on a regular schedule, not just once.

#289 · Laboratory Equipment and Supplies

Why must reagents and consumables be ordered and received according to set inventory procedures?

Answer

Following inventory procedures prevents the lab from running out of critical supplies and ensures items are logged, dated, and rotated correctly. Poor inventory control can lead to expired reagents being used unknowingly.

#290 · Laboratory Equipment and Supplies

Why do temperature requirements matter when storing reagents and consumables?

Answer

Many reagents are only stable within a specific temperature range, so storing them outside that range can degrade them and cause inaccurate test results. The MLA/T must check and follow the storage temperature stated for each item on receipt and in ongoing storage.

#291 · Laboratory Equipment and Supplies

What is a centrifuge used for in the lab, and what mandatory equipment variant is listed alongside it?

Answer

A centrifuge spins specimens at speed to separate components by density, such as separating serum from cells. A microhematocrit centrifuge is a separate mandatory piece of equipment used specifically for hematocrit testing on small capillary samples.

#292 · Laboratory Equipment and Supplies

What is the difference between a mechanical balance and an analytical balance?

Answer

Both are mandatory equipment used to weigh substances in the lab. An analytical balance is built for much higher precision than a mechanical balance, so it is used when very small, exact weights are required.

#293 · Laboratory Equipment and Supplies

Why must refrigerators and freezers in the lab be monitored?

Answer

Refrigerators and freezers store reagents, specimens, and controls that require specific temperature ranges to remain stable. Regular monitoring, including with a thermometer, confirms the unit is holding the correct temperature so stored items are not compromised.

#294 · Laboratory Equipment and Supplies

What extra feature does the optional refrigerator/freezer with alarm system and recorder add over a standard unit?

Answer

It automatically records temperature over time and sounds an alarm if the temperature drifts outside the safe range. This gives continuous documentation and an early warning, rather than relying only on periodic manual checks.

#295 · Laboratory Equipment and Supplies

What is an autoclave used for in the laboratory?

Answer

An autoclave uses pressurized steam to sterilize equipment, media, and waste. It is mandatory equipment because proper sterilization is essential to prevent contamination and control biohazards.

#296 · Laboratory Equipment and Supplies

What is the purpose of a loop incinerator, bacti-cinerator, or disposable loop in microbiology work?

Answer

These are used to sterilize an inoculating loop between uses, or to provide a single-use sterile loop, preventing cross-contamination between specimens or culture plates. Proper use protects both sample integrity and the safety of the person performing the work.

#297 · Laboratory Equipment and Supplies

What safety equipment must be available for handling spills and chemical exposure in the lab?

Answer

Mandatory items include a spill kit, an eyewash station, and a flammable cabinet, along with PPE such as gloves, goggles, splash shields, lab coats, and gowns. These items must be accessible and staff must know how to use them correctly in an emergency.

#298 · Laboratory Equipment and Supplies

What is a Class 1 Biological Safety Cabinet used for, and how does it differ from Class II and III cabinets?

Answer

A Class 1 Biological Safety Cabinet is mandatory equipment that protects the worker and environment but not necessarily the specimen from contamination. Class II and III cabinets are optional equipment requiring theory knowledge only, and they provide increasingly higher levels of specimen and worker protection.

#299 · Laboratory Equipment and Supplies

What are mixing devices such as a vortex or shaker used for in the lab?

Answer

These devices resuspend settled particles, mix reagents with samples, or blend solutions evenly before testing. Using them correctly ensures a specimen or reagent is homogeneous, which is necessary for an accurate result.

#300 · Laboratory Equipment and Supplies

What is a waterbath used for, and what related device measures its accuracy?

Answer

A waterbath maintains samples or reagents at a constant, controlled temperature for procedures that require it. A thermometer, also mandatory equipment, is used to verify the waterbath is actually holding the correct temperature.

#301 · Histology and Cytology

What is fixation in histology?

Answer

Fixation is the process of preserving tissue with a chemical agent, most commonly formalin, right after it leaves the body. It stops autolysis and bacterial decay and hardens the tissue so it keeps its structure through later processing steps.

#302 · Histology and Cytology

What is decalcification and when is it needed?

Answer

Decalcification is the removal of calcium salts from bone or heavily calcified tissue using an acid or chelating solution. It must be done before the tissue can be sectioned, because a microtome blade cannot cut cleanly through hard calcified material.

#303 · Histology and Cytology

What are the main steps of tissue processing in histology?

Answer

Tissue processing moves fixed tissue through dehydration in graded alcohols, clearing in a solvent such as xylene, and infiltration with paraffin wax. This removes water and replaces it with a medium firm enough to allow thin, even sectioning.

#304 · Histology and Cytology

What is the purpose of cytology fixation and preservation?

Answer

Cytology fixation and preservation stop cell degeneration and keep the cellular detail intact once a specimen is collected. Common methods include alcohol-based fixative sprays or immersion, chosen based on the specimen type and the stain that will follow.

#305 · Histology and Cytology

Why is centrifugation used in cytology specimen processing?

Answer

Centrifugation concentrates cells from a fluid specimen, such as urine or a body cavity fluid, by spinning them to the bottom of the tube. The concentrated cell pellet is then used to make a smear or cell block so scarce cells are not missed on review.

#306 · Histology and Cytology

What is a cell block preparation?

Answer

A cell block is made by concentrating cells from a fluid specimen and processing them like a small tissue sample, embedding them in paraffin. It lets the lab cut multiple thin sections and perform special stains, which is not possible from a simple smear.

#307 · Histology and Cytology

What is cytocentrifugation used for?

Answer

Cytocentrifugation spins a diluted fluid specimen at low speed directly onto a slide, depositing a thin monolayer of cells in a small circular area. This concentrates cells while keeping them well spread and undistorted for easier microscopic review.

#308 · Histology and Cytology

How does a direct smear preparation differ from liquid-based processing in cytology?

Answer

A direct smear is made by spreading the specimen straight onto a slide by hand, which is fast but can result in uneven thickness or obscuring debris. Liquid-based processors first suspend the cells in a preservative fluid, then filter or spin a cleaner, more uniform layer onto the slide.

#309 · Histology and Cytology

What does Hematoxylin and Eosin (H&E;) stain show on a tissue section?

Answer

Hematoxylin stains cell nuclei blue to purple because it binds acidic structures like DNA, while eosin stains cytoplasm and connective tissue pink to red because it binds basic structures. Together they are the standard stain that lets pathologists see overall tissue architecture.

#310 · Histology and Cytology

What is Romanowsky staining used for?

Answer

Romanowsky-type stains, such as Wright or Giemsa stains, use a mix of dyes to color blood and bone marrow cell components in contrasting shades. They are the standard stains for examining blood smears and differentiating white blood cell types.

#311 · Histology and Cytology

What is the Papanicolaou (Pap) stain used for?

Answer

The Papanicolaou stain is a multichromatic stain designed for cytology specimens, most famously cervical Pap smears. It highlights fine nuclear detail and differentiates cytoplasm maturity, which helps identify precancerous or abnormal cells.

#312 · Histology and Cytology

Why must stains and reagents in histology and cytology be prepared and maintained carefully?

Answer

Stains and reagents degrade over time or with contamination, which can weaken color intensity or produce unreliable results. Following expiry dates, proper storage, and quality checks on freshly prepared reagents keeps staining results consistent and diagnostically reliable.

#313 · Histology and Cytology

Why is cross-contamination a major risk in histology and cytology processing?

Answer

Small fragments of tissue or cells from one specimen can transfer onto another during cutting, staining, or handling, which can create a false positive or wrong diagnosis for a patient. Cleaning instruments between specimens, using fresh reagents, and following strict handling order all reduce this risk.

#314 · Histology and Cytology

What steps help minimize cross-contamination risk during specimen processing?

Answer

Processing one specimen at a time where possible, cleaning blades and forceps between cases, changing staining solutions on a schedule, and carefully labeling containers all help prevent tissue or cell carryover. Staff must also watch for floaters, which are stray tissue fragments that land on the wrong slide.

#315 · Histology and Cytology

What is the difference between manual and automated staining in histology and cytology?

Answer

Manual staining involves moving slides by hand through a series of staining and rinsing dishes, giving flexibility but more variability. Automated stainers move slides through the same steps on a programmed timer, which improves consistency and frees up staff time.

#316 · Histology and Cytology

What is coverslipping and why is it done?

Answer

Coverslipping is applying a thin glass or plastic cover over the stained tissue or cell sample using a mounting medium. It protects the specimen from damage and creates a flat, clear optical surface so the slide can be viewed properly under a microscope.

#317 · Histology and Cytology

Name two quality assurance requirements relevant to histology and cytology.

Answer

Quality assurance includes running control slides with known results alongside patient stains to confirm the stain worked correctly, and checking that specimen and slide labels match at every step. Documented monitoring of reagents, equipment temperatures, and staining times is also part of ongoing quality assurance.

#318 · Histology and Cytology

Why is formalin a laboratory safety hazard in histology?

Answer

Formalin is a toxic, irritating chemical fixative that can cause respiratory irritation, skin and eye damage, and is a known carcinogen with prolonged exposure. It must be handled in a fume hood or with proper ventilation, and staff need appropriate personal protective equipment.

#319 · Histology and Cytology

Why is xylene a laboratory safety hazard in histology?

Answer

Xylene is a flammable, volatile solvent used in tissue clearing and coverslipping that can cause respiratory and skin irritation with repeated exposure. It should be used in a well-ventilated area or fume hood, kept away from ignition sources, and handled with gloves.

#320 · Histology and Cytology

What safety practices apply specifically to histology and cytology work areas?

Answer

Staff should work with formalin and xylene under proper ventilation or in a fume hood, wear gloves and eye protection, and know the location of safety data sheets and spill kits. Waste chemicals must be disposed of according to hazardous waste procedures rather than poured down a sink.

#321 · Histology and Cytology

What information must be tracked when filing and storing histology blocks and slides?

Answer

Each paraffin block and slide must be labeled with a unique identifier linking it back to the patient specimen and accession number. Accurate filing systems let the lab retrieve blocks or slides quickly for additional cuts, special stains, or later comparison.

#322 · Histology and Cytology

Why are histology blocks and slides retained for extended periods rather than discarded quickly?

Answer

Retained blocks and slides may be needed later for additional testing, a second pathologist opinion, or comparison if the patient develops related findings in the future. Following the facility's retention policy ensures this material is available when needed.

#323 · Histology and Cytology

What is the general purpose of tissue and specimen retrieval procedures in histology and cytology?

Answer

Retrieval procedures make sure a stored block, slide, or specimen can be located accurately and efficiently using its identifier and storage location. This supports timely re-testing, quality checks, and legal or accreditation record-keeping requirements.

#324 · Histology and Cytology

How does specimen processing differ overall between histology and cytology?

Answer

Histology processes solid tissue samples through fixation, decalcification if needed, and paraffin embedding before thin sectioning. Cytology processes loose cells from fluids or scrapings through methods like centrifugation, cytocentrifugation, or direct smearing, since there is no solid tissue block to section.

#325 · Histology and Cytology

Why must a Medical Laboratory Assistant/Technician understand both histology and cytology procedures even in a limited scope role?

Answer

Even when not performing every step personally, understanding fixation, processing, staining, and safety procedures helps the MLA/T handle specimens correctly at every stage they do touch. This protects specimen integrity and prevents errors that could affect a patient's diagnosis.

#326 · Clinical Microbiology

List the major classifications of microorganisms covered in clinical microbiology.

Answer

The major classifications include bacteria, viruses, parasites, protozoa, fungi (including molds and yeasts), chlamydia, and rickettsia. Each group has different growth requirements, structures, and detection methods in the clinical lab.

#327 · Clinical Microbiology

What is normal flora?

Answer

Normal flora are the microorganisms that routinely live on or in a healthy person's body, such as on the skin or in the gut, without causing disease. They can even provide benefits, like helping with digestion or preventing harmful organisms from establishing themselves.

#328 · Clinical Microbiology

What is an opportunist organism in microbiology?

Answer

An opportunist is a microorganism that does not normally cause disease in a healthy host but can cause infection when the host's defenses are weakened, such as during illness or immune suppression. It takes advantage of a change in the body's normal balance to become harmful.

#329 · Clinical Microbiology

What does the term commensal mean in microbiology?

Answer

A commensal organism lives on or in a host and benefits from that relationship without helping or harming the host under normal conditions. Many normal flora organisms are considered commensals.

#330 · Clinical Microbiology

What is a pathogen?

Answer

A pathogen is a microorganism capable of causing disease in its host. Its ability to cause illness may depend on factors like the organism's virulence, the number of organisms present, and the host's immune status.

#331 · Clinical Microbiology

What do Risk Group classifications (1 through 4) indicate for microorganisms?

Answer

Risk Groups classify organisms by the level of hazard they pose to laboratory workers and the community, from Risk Group 1, which poses little to no risk, up to Risk Group 4, which causes severe disease with high risk of spread and no available treatment. These classifications guide the biosafety precautions required when handling each organism.

#332 · Clinical Microbiology

What general requirements apply to transporting microbiology specimens?

Answer

Microbiology specimens must be transported promptly in appropriate containers or transport media to keep organisms viable and prevent overgrowth of contaminants. Specimens must be properly sealed, labeled, and kept at the correct temperature, since delays or improper handling can cause false negative results.

#333 · Clinical Microbiology

What is a non-selective growth medium?

Answer

A non-selective medium supports the growth of a wide range of microorganisms without inhibiting any particular type. Blood agar is a common example, allowing most bacteria found in a specimen to grow.

#334 · Clinical Microbiology

What is a selective growth medium?

Answer

A selective medium contains ingredients that inhibit the growth of certain organisms while allowing others to grow, helping isolate a specific type of organism from a mixed specimen. MacConkey agar, for example, inhibits most gram-positive bacteria while allowing gram-negative bacteria to grow.

#335 · Clinical Microbiology

What is a differential growth medium?

Answer

A differential medium allows different types of organisms to grow but shows a visible difference between them, such as a color change, based on how each organism reacts with the medium's ingredients. This helps distinguish between organisms based on their biochemical activity.

#336 · Clinical Microbiology

What is an enriched growth medium?

Answer

An enriched medium contains extra nutrients, such as blood or serum, added to support the growth of fastidious organisms that would not grow well on standard media. Chocolate agar is an example, providing nutrients that certain bacteria need to survive.

#337 · Clinical Microbiology

What is Blood Agar used for?

Answer

Blood Agar is a non-selective, enriched medium containing sheep's blood that supports the growth of many types of bacteria. It also allows technicians to observe hemolysis patterns, which help identify certain bacterial species.

#338 · Clinical Microbiology

What is MacConkey Agar used for?

Answer

MacConkey Agar is a selective and differential medium that inhibits gram-positive organisms and allows gram-negative bacteria to grow. It also differentiates lactose-fermenting from non-lactose-fermenting organisms by a color change, which helps identify gram-negative bacteria such as those in the gut.

#339 · Clinical Microbiology

What is Chocolate Agar used for?

Answer

Chocolate Agar is an enriched medium made by heating blood agar, which releases nutrients that support fastidious organisms like *Haemophilus* and *Neisseria* species. It is named for its brown color, not for containing chocolate.

#340 · Clinical Microbiology

What is CNA agar used for?

Answer

CNA agar is a selective medium containing colistin and nalidixic acid, which inhibit most gram-negative bacteria while allowing gram-positive organisms to grow. It is useful for isolating gram-positive bacteria from specimens with mixed bacterial flora.

#341 · Clinical Microbiology

What is GC selective media used for?

Answer

GC selective media is designed to isolate *Neisseria gonorrhoeae* and related organisms by inhibiting the normal flora and contaminating organisms commonly found in genital specimens. It typically includes antibiotics that suppress unwanted competing bacteria.

#342 · Clinical Microbiology

What is the streak plate method used for in specimen inoculation?

Answer

The streak plate method spreads a specimen across an agar plate in a pattern of decreasing density, usually across several quadrants, to isolate single colonies of bacteria. Isolated colonies let the lab identify and test individual organisms rather than a mixed clump.

#343 · Clinical Microbiology

What is an agar slant tube used for?

Answer

An agar slant tube holds solid medium poured at an angle in a test tube, providing a larger sloped surface area for culturing and maintaining bacterial isolates. It is often used to keep a pure culture viable for storage or later testing.

#344 · Clinical Microbiology

What is a culture broth tube used for?

Answer

A culture broth tube contains liquid growth medium used to grow bacteria in suspension rather than on a solid surface. Broths can help detect small numbers of organisms or support the growth of organisms that need liquid conditions.

#345 · Clinical Microbiology

What is an automated plate streaker used for?

Answer

An automated plate streaker is an instrument that mechanically inoculates and streaks specimens onto agar plates in a consistent pattern. It improves standardization and saves technologist time compared to manual streaking.

#346 · Clinical Microbiology

Why must slides for microbiological review be prepared carefully?

Answer

Slides must be prepared with an even, appropriately thin smear of the specimen so organisms can be clearly seen and correctly identified under the microscope. Poorly prepared slides can hide organisms or distort their appearance, leading to misidentification.

#347 · Clinical Microbiology

Why are correct time and temperature requirements important during incubation of microbiology specimens?

Answer

Different organisms grow best at specific temperatures and need a minimum incubation time to form visible, identifiable colonies. Using the wrong temperature or stopping incubation too early can cause slow-growing or fastidious organisms to be missed.

#348 · Clinical Microbiology

What are anaerobic incubation conditions used for, and how are they created?

Answer

Anaerobic conditions are used to grow organisms that cannot survive in the presence of oxygen. They can be created using an anaerobic jar with a gas-generating pack, known as the gas-pak method, or with an anaerobic glove box that maintains an oxygen-free environment for handling and incubating specimens.

#349 · Clinical Microbiology

What is increased CO₂ tension incubation used for?

Answer

Increased CO₂ tension incubation, provided by a CO₂ incubator, supplies a higher concentration of carbon dioxide that certain organisms, such as *Neisseria* and *Haemophilus* species, need for optimal growth. Without this enriched atmosphere, these organisms may grow poorly or not at all.

#350 · Clinical Microbiology

What are microaerophilic conditions?

Answer

Microaerophilic conditions provide a reduced oxygen level, lower than normal atmospheric oxygen but higher than fully anaerobic conditions. Organisms like *Campylobacter* species require this specific low-oxygen environment to grow successfully.

#351 · Clinical Microbiology

What is the basic function of automated microbial identification and susceptibility testing systems?

Answer

These automated systems analyze bacterial growth patterns or biochemical reactions to identify the organism species and determine which antibiotics will effectively treat it. They speed up results and improve standardization compared to fully manual identification and testing methods.

#352 · Clinical Microbiology

What is the basic function of automated blood culture systems?

Answer

Automated blood culture systems continuously monitor inoculated blood culture bottles for signs of microbial growth, such as gas production or changes detected by sensors. When growth is detected, the system flags the bottle so staff can perform further identification testing promptly.

#353 · Clinical Microbiology

What general lab procedures apply to preparing culture media?

Answer

Preparing media involves accurately measuring and dissolving the correct ingredients, sterilizing the mixture, pouring it into plates or tubes under sterile conditions, and allowing it to solidify or cool properly. Quality checks on the finished media confirm sterility and expected performance before it is used with patient specimens.

#354 · Clinical Microbiology

What is the principle behind the Gram stain?

Answer

The Gram stain uses a crystal violet dye followed by an iodine mordant, then a decolorizing step, and finally a safranin counterstain. Gram-positive bacteria retain the crystal violet and appear purple because of their thick peptidoglycan cell wall, while gram-negative bacteria lose the crystal violet during decolorization and appear pink from the safranin counterstain.

#355 · Clinical Microbiology

What is the procedure for performing a Gram stain?

Answer

A smear is fixed to a slide, then flooded with crystal violet, rinsed, treated with iodine, rinsed, decolorized briefly with alcohol or acetone, rinsed again, and finally counterstained with safranin before a final rinse. The slide is then examined under the microscope to determine the Gram reaction and cell shape.

#356 · Clinical Microbiology

What is the principle behind acid-fast staining?

Answer

Acid-fast staining identifies organisms, such as *Mycobacterium* species, that have a waxy, lipid-rich cell wall resistant to normal decolorization. These organisms retain the primary stain even after treatment with an acid-alcohol decolorizer, appearing a distinct color while other organisms lose the stain and take up the counterstain.

#357 · Clinical Microbiology

What is the principle of fluorescent stains in microbiology?

Answer

Fluorescent stains bind to specific structures within or on microorganisms and glow under ultraviolet or specific wavelength light, making the organisms easier to spot against a dark background. This can improve detection speed and sensitivity compared to standard light microscopy stains.

#358 · Clinical Microbiology

What is the principle of fluorescent antibody staining?

Answer

Fluorescent antibody staining uses antibodies labeled with a fluorescent dye that bind specifically to a target organism or antigen. When viewed under fluorescent microscopy, only the organisms bound by the labeled antibody glow, allowing specific identification of that organism in a mixed sample.

#359 · Clinical Microbiology

What is the fundamental structural difference between gram-positive and gram-negative bacteria?

Answer

Gram-positive bacteria have a thick peptidoglycan cell wall that retains crystal violet dye during Gram staining, making them appear purple. Gram-negative bacteria have a thinner peptidoglycan layer plus an outer membrane, which does not retain the crystal violet after decolorization, making them appear pink from the counterstain.

#360 · Clinical Microbiology

Why does the outer membrane of gram-negative bacteria matter clinically?

Answer

The outer membrane of gram-negative bacteria contains lipopolysaccharide, which can trigger a strong immune and inflammatory response in the host and contributes to some gram-negative infections being more severe. It also affects how certain antibiotics can penetrate the cell.

#361 · Clinical Microbiology

What is the difference between cocci and bacilli bacteria?

Answer

Cocci are round or spherical shaped bacteria, which can appear singly, in pairs, chains, or clusters depending on the species. Bacilli are rod shaped bacteria, and their length, width, and end shape can help with identification.

#362 · Clinical Microbiology

Why is bacterial shape (cocci versus bacilli) useful information in microbiology?

Answer

Observing whether an organism is round or rod shaped under the microscope, combined with its Gram stain reaction, gives an early clue to help narrow down likely organism groups before further testing. This guides which additional tests or media the lab uses next.

#363 · Clinical Microbiology

How do fungi, molds, and yeasts differ as a microorganism classification from bacteria?

Answer

Fungi, molds, and yeasts are eukaryotic organisms with a true nucleus and more complex cell structure than bacteria, which are prokaryotic. They often require different growth media, incubation conditions, and staining or identification methods than bacteria do.

#364 · Clinical Microbiology

What distinguishes chlamydia and rickettsia from typical free-living bacteria?

Answer

Chlamydia and rickettsia are types of bacteria that cannot grow on standard culture media because they require a living host cell to reproduce. This obligate intracellular lifestyle means they are typically detected using molecular, antigen, or serologic testing rather than routine culture.

#365 · Clinical Microbiology

Why does a Medical Laboratory Assistant/Technician need to understand parasites and protozoa as separate categories from bacteria and viruses?

Answer

Parasites and protozoa are usually larger, more complex organisms that may require different specimen types, such as stool or blood, and different detection methods like microscopic examination for ova and parasites rather than culture. Recognizing these differences helps ensure the right collection, processing, and testing approach is used for each specimen.

#366 · Clinical Chemistry

What is a reportable range in clinical chemistry?

Answer

It is the span of test result values that a lab can accurately report, based on the analyzer's validated linear range. Results outside this range need dilution or another method before they can be reported. An MLA/T must recognize when a result falls outside the reportable range so it is not sent out as a final number.

#367 · Clinical Chemistry

What is a variant result in chemistry testing?

Answer

A variant result is one that falls outside the expected or usual pattern for a test, which may signal an instrument problem, a specimen problem, or a true critical patient value. Recognizing variant results lets staff flag them for review before release. This protects patients from a wrong or misleading report.

#368 · Clinical Chemistry

What does a lipemic sample look like and why does it matter?

Answer

A lipemic sample is cloudy or milky because of high fat content in the blood, usually from triglycerides. Lipemia can interfere with many chemistry assays by scattering light or blocking the analyzer's optical reading. The lab must recognize lipemia and take the appropriate action, such as flagging or correcting the result.

#369 · Clinical Chemistry

What does a hemolyzed sample look like and why does it matter?

Answer

A hemolyzed sample has a pink or red tinge in the serum or plasma because red blood cells have broken open and released their contents. This falsely raises results like potassium and certain enzymes because those substances are concentrated inside red cells. The sample often needs to be flagged or redrawn depending on the test.

#370 · Clinical Chemistry

What does an icteric sample look like and why does it matter?

Answer

An icteric sample has a yellow to brown color caused by high bilirubin levels. Icterus can interfere with the optical measurement of some chemistry tests, causing inaccurate results. Staff need to recognize icterus and take appropriate action before the result is reported.

#371 · Clinical Chemistry

Why are some chemistry specimens described as time-dependent for analysis?

Answer

Certain analytes break down or change over time once collected, so the specimen must be processed within a specific window. Urine is a common example, since components like glucose or bilirubin can degrade if the sample sits too long. Recognizing a time-dependent sample means acting quickly rather than letting it wait on the bench.

#372 · Clinical Chemistry

What is included in routine and microscopic (R&M;) urinalysis?

Answer

R&M; urinalysis includes a physical and chemical screen of the urine plus a microscopic look at the sediment for cells, casts, crystals, and other formed elements. The MLA/T is responsible for proper specimen collection, deciding whether to reject an unsuitable sample, preparing the specimen, and preserving it for accurate microscopic examination.

#373 • Clinical Chemistry

Give one reason a urine specimen might be rejected for R&M; testing.

Answer

A urine specimen can be rejected if it is not properly labeled, was collected in the wrong container, has sat too long without preservation, or is grossly contaminated. Rejecting an unsuitable specimen prevents an inaccurate microscopic or chemical result from reaching the patient's chart.

#374 • Clinical Chemistry

How is total volume measured for a 24-hour urine collection?

Answer

The patient collects all urine passed over a full 24-hour period into one large container, and the lab measures the total volume once collection is complete. This total volume is then used to calculate excretion rates for substances like creatinine or protein. Preservation guidelines, such as refrigeration or a chemical preservative, must be followed throughout the collection.

#375 · Clinical Chemistry

Why do 24-hour urine collections require preservation?

Answer

Without preservation, components in urine can break down, precipitate, or be affected by bacterial growth over the long collection period. Following the lab's preservation guidelines, such as keeping the container cold or adding a preservative, keeps the analytes stable until testing. Improper preservation can lead to falsely low or inaccurate results.

#376 · Clinical Chemistry

What is glucose tolerance testing used to evaluate?

Answer

Glucose tolerance testing measures how the body processes a glucose load over time, often used to screen for diabetes or gestational diabetes. The patient drinks a glucose solution and blood is drawn at timed intervals afterward. The MLA/T follows the lab procedure for administering the drink and collecting each timed specimen.

#377 • Clinical Chemistry

What should staff do if a patient reacts negatively during a glucose load test?

Answer

Staff must recognize signs of a negative reaction, such as nausea, dizziness, or vomiting, and initiate the lab's protocol for ending the test safely. Patient safety takes priority over completing the timed collection. The event should be documented according to lab procedure.

#378 • Clinical Chemistry

What is timed specimen collection in tolerance testing?

Answer

Timed specimen collection means drawing blood samples at specific, pre-set intervals after the patient consumes the glucose solution, such as fasting, then 1 hour, then 2 hours. Accurate timing is essential because the test results depend on comparing glucose levels at each exact interval. Missing or mistiming a draw can invalidate the test.

#379 · Clinical Chemistry

What does a liver function panel generally assess?

Answer

A liver function panel groups tests that evaluate how well the liver is working, often including enzymes and proteins related to liver cell damage or bile flow. Grouping these tests together gives a fuller picture of liver health than any single test alone. Recognizing why this panel is ordered helps staff understand its clinical purpose.

#380 · Clinical Chemistry

What does a renal function panel generally assess?

Answer

A renal function panel groups tests that evaluate how well the kidneys are filtering and balancing the blood, often including markers related to waste clearance. It is ordered when kidney disease or a change in kidney function is suspected. Understanding the rationale for grouped renal tests helps explain why they are ordered together rather than individually.

#381 · Clinical Chemistry

What is a lipid profile used for?

Answer

A lipid profile is a grouped chemistry panel used to assess a patient's fat-related blood components, which relate to cardiovascular risk. It is commonly ordered as part of routine health screening or heart disease risk assessment. Grouping these tests together gives clinicians a fuller picture than any single lipid value.

#382 · Clinical Chemistry

Why are cardiac markers ordered as a group?

Answer

Cardiac markers are grouped because they help clinicians assess whether a patient may be having a heart-related event, such as damage to heart muscle. Looking at several markers together gives more diagnostic confidence than a single test. This is one of the grouped chemistry panels an MLA/T should be able to identify and explain.

#383 · Clinical Chemistry

What is the purpose of grouped endocrine function testing?

Answer

Endocrine function panels group tests that evaluate hormone-related conditions, helping clinicians assess how glands are functioning. Ordering these tests together supports diagnosis and monitoring of endocrine disorders. Understanding the rationale for this grouping is part of recognizing why chemistry panels exist.

#384 · Clinical Chemistry

What are tumour markers used for in chemistry testing?

Answer

Tumour markers are substances measured in blood that can be associated with the presence or monitoring of certain cancers. They are grouped as their own category of chemistry testing because of their specific clinical purpose. An MLA/T should be able to identify this as one of the grouped test categories, even without interpreting results.

#385 · Clinical Chemistry

Why is drug level testing grouped separately in chemistry?

Answer

Drug level testing measures the concentration of specific medications in a patient's blood, often to monitor for effective dosing or to avoid toxicity. It is treated as its own grouped category because it serves a therapeutic monitoring purpose distinct from general chemistry screening. Recognizing this rationale helps explain why it is ordered on its own.

#386 · Clinical Chemistry

What is hepatitis testing checking for in a chemistry panel?

Answer

Hepatitis testing is grouped to screen for or diagnose viral hepatitis infections affecting the liver. It is distinct from general liver function testing because it targets markers specific to hepatitis viruses. This is one of the grouped chemistry test categories an MLA/T should recognize.

#387 · Clinical Chemistry

What is included in electrolyte testing?

Answer

Electrolyte testing is a grouped chemistry panel that measures charged particles in the blood important for fluid balance, nerve, and muscle function. It is one of the most commonly ordered grouped panels because electrolyte imbalance can affect many body systems. Understanding why it is grouped helps explain its frequent clinical use.

#388 · Clinical Chemistry

What is the rationale for ordering glucose testing?

Answer

Glucose testing is grouped as its own category because blood sugar levels are central to diagnosing and monitoring conditions like diabetes. It can be ordered as a stand-alone test or as part of tolerance testing. This is one of the grouped chemistry test categories an MLA/T should be able to describe.

#389 · Clinical Chemistry

What is the basic principle behind an automated biochemistry analyzer?

Answer

An automated biochemistry analyzer processes patient specimens through a series of steps such as sample aspiration, reagent addition, and reaction measurement, often using light-based detection. This automation lets the lab run many tests accurately and consistently at high volume. Understanding this principle helps staff recognize normal analyzer operation versus a malfunction.

#390 · Clinical Chemistry

Why is understanding automated analyzer operation part of MLA/T competency?

Answer

MLA/Ts need to describe the principles of how automated biochemistry analyzers work so they can recognize when results or instrument behavior look wrong. This supports troubleshooting and quality control in the daily workflow. It also helps staff understand how sample interferences like lipemia or hemolysis affect the automated reading.

#391 • Clinical Chemistry

What action should be taken with a lipemic, hemolyzed, or icteric sample before releasing chemistry results?

Answer

The sample should be recognized as compromised and handled per lab procedure, which may include flagging the result, running an interference correction, or requesting a redraw. This category of sample recognition is a hands-on, non-simulation competency, meaning it must be demonstrated with real specimens. It protects patients from inaccurate reported results.

#392 • Clinical Chemistry

Why can up to 70% of clinical chemistry competencies be assessed in simulation, but not all?

Answer

Many chemistry skills like recognizing reportable ranges, performing urinalysis steps, or describing analyzer principles can be taught and tested through simulated scenarios. However, recognizing real sample defects like lipemia, hemolysis, and icterus requires actual physical specimens, so that competency cannot be simulated. This distinction matters for how MLA/T training and testing are structured.

#393 · Clinical Chemistry

What is the difference between reportable range and variant result?

Answer

Reportable range is the numeric span an analyzer can validly report, while a variant result is a value that deviates from the expected or typical pattern for that patient or test. A result can be within the reportable range and still be flagged as a variant if it is unusual. Both concepts require the MLA/T to pause and evaluate before releasing results.

#394 · Clinical Chemistry

Why must microscopic urine samples be preserved properly before examination?

Answer

Formed elements in urine, such as cells and casts, can break down or dissolve if the sample sits too long without preservation. Proper preservation keeps the specimen stable so the microscopic exam reflects what was actually present in the patient. This is part of the R&M; urinalysis competency covering preparation and preservation.

#395 · Clinical Chemistry

What is meant by specimen rejection in urinalysis?

Answer

Specimen rejection means the lab determines a urine sample is not suitable for testing and will not proceed with analysis until a new sample is provided. Reasons can include improper collection, inadequate volume, or an unpreserved sample that sat too long. This decision is part of the MLA/T's specimen collection and rejection responsibilities.

#396 · Clinical Chemistry

What role does the MLA/T play during glucose solution administration?

Answer

The MLA/T provides the glucose solution to the patient as part of tolerance and stimulation testing, following the lab's exact procedure for the dose and timing. They also monitor the patient afterward for reactions and manage the timed specimen collection that follows. This combines a patient-facing role with a technical collection role.

#397 · Clinical Chemistry

Name two examples of grouped chemistry tests in the MLA/T competency profile.

Answer

Examples include liver function tests and renal function tests, which are each grouped together because they give a fuller clinical picture than a single isolated test. Other groups include lipid profile, cardiac markers, and electrolytes. MLA/Ts are expected to identify these groupings and describe the rationale behind ordering them together.

#398 · Clinical Chemistry

Why is recognizing a hemolyzed sample especially important for potassium testing?

Answer

Red blood cells contain much higher potassium concentrations than plasma, so when cells rupture during hemolysis, potassium leaks out and falsely raises the measured level. This can create a misleading result that does not reflect the patient's true potassium status. Staff must flag hemolyzed samples so clinicians are not misled by a falsely elevated result.

#399 · Clinical Chemistry

What does stimulation testing refer to alongside tolerance testing?

Answer

Stimulation testing involves giving a substance to trigger a measurable body response, similar in structure to tolerance testing where a glucose load triggers a measurable glucose response. Both types of testing require the lab to follow a defined procedure, monitor the patient, and collect specimens at set times. The MLA/T supports both types by following the lab's specific protocol.

#400 · Clinical Chemistry

Why is describing the types of routine tolerance and glucose load testing part of MLA/T competency?

Answer

Understanding the different types of tolerance and glucose load tests helps the MLA/T follow the correct protocol for each specific test ordered, since procedures and timing can vary. This knowledge supports accurate specimen collection at the right intervals. It also helps staff recognize what a normal versus abnormal testing process looks like.

#401 · Clinical Chemistry

What is the clinical value of ordering an electrolyte panel as a group rather than single tests?

Answer

Electrolytes work together in the body to maintain fluid and acid-base balance, so viewing them as a group gives a clearer picture of a patient's overall balance than any one value alone. An imbalance in one electrolyte often relates to changes in another. This is why electrolyte testing is treated as a grouped chemistry panel.

#402 · Clinical Chemistry

How does time-dependence affect urine testing differently than blood chemistry testing?

Answer

Urine components such as glucose, bilirubin, and formed elements can degrade or change more quickly at room temperature than many blood analytes, making prompt processing especially important. This is why urine is specifically named as an example of a time-dependent specimen. Delays can lead to false or altered results in urinalysis.

#403 · Clinical Chemistry

Why can chemistry analyzer principles be assessed through simulation?

Answer

Describing how an automated biochemistry analyzer operates, such as its general steps of sample handling, reagent mixing, and detection, is a knowledge-based competency that can be tested through scenarios or written assessment. This differs from hands-on recognition of a physical sample defect, which needs a real specimen. That is why analyzer principles fall under the simulation-eligible portion of the category.

#404 · Clinical Chemistry

What is a key safety consideration during a glucose tolerance test?

Answer

Staff must watch for negative patient reactions during the test, such as feeling faint or unwell after drinking the glucose solution, and be ready to stop the test following the lab's protocol. Patient wellbeing always comes before completing every timed draw. This safety monitoring is a defined part of the tolerance testing competency.

#405 · Clinical Chemistry

Why might a lab order both liver function and hepatitis testing on the same patient?

Answer

Liver function tests show how well the liver is currently working, while hepatitis testing looks specifically for viral infection markers that could be causing liver damage. Ordering both together helps determine whether abnormal liver function is linked to a hepatitis infection. They are treated as two separate grouped panels because they serve different diagnostic purposes.

#406 · Clinical Chemistry

What should an MLA/T do upon noticing a chemistry result appears to be a variant result?

Answer

The MLA/T should recognize that the result falls outside the expected pattern and flag it for further review before it is released, rather than reporting it automatically. This may involve rechecking the specimen for interference or repeating the test. Careful review of variant results helps prevent inaccurate patient reports.

#407 · Clinical Chemistry

Why does urine collection for R&M; require attention to preparation, not just collection?

Answer

Even a properly collected urine sample needs correct preparation, such as adequate mixing before viewing the sediment, to give an accurate microscopic result. Skipping proper preparation can cause formed elements to be missed or unevenly distributed. This is why preparation is listed as its own step alongside collection and preservation.

#408 · Clinical Chemistry

What distinguishes cardiac marker testing from a general chemistry panel?

Answer

Cardiac markers are a specific grouped set of tests aimed at detecting possible heart muscle injury or cardiac events, rather than giving a broad overview of general body chemistry. They are ordered when a cardiac event is suspected, often in urgent situations. This focused clinical purpose is why they are their own grouped category.

#409 · Clinical Chemistry

Why is it important for an MLA/T to know the rationale behind each grouped chemistry panel, not just the test names?

Answer

Knowing the rationale helps the MLA/T understand why certain tests are ordered together and recognize whether a panel makes clinical sense for a given patient scenario. This deeper understanding supports better specimen handling and communication with the healthcare team. It goes beyond memorizing panel names to understanding their clinical purpose.

#410 · Clinical Chemistry

Summarize why sample quality recognition, covering lipemic, hemolyzed, icteric, and time-dependent samples, is a core clinical chemistry competency.

Answer

Sample quality issues can distort chemistry results in specific, predictable ways, so recognizing them protects the accuracy of patient reports. This competency requires hands-on judgment with real specimens rather than only simulation. It directly connects specimen handling skills to patient safety in chemistry testing.

#411 · Clinical Hematology

What does a complete blood count (CBC) measure?

Answer

A CBC measures the main cellular components of blood, including hemoglobin, hematocrit, red blood cell count and indices, white blood cell count, platelets, and the differential of white cell types. Each component has its own normal range that helps assess a patient's overall blood health. Describing these components and ranges is a core hematology competency.

#412 · Clinical Hematology

What does hemoglobin measure in a CBC?

Answer

Hemoglobin measures the protein in red blood cells responsible for carrying oxygen throughout the body. It is one of the core components of the CBC and has its own defined normal range. Low hemoglobin can indicate anemia, while unusually high levels can also be clinically significant.

#413 · Clinical Hematology

What does hematocrit measure in a CBC?

Answer

Hematocrit measures the proportion of whole blood volume that is made up of red blood cells, expressed as a percentage. It generally rises and falls together with hemoglobin since both reflect red cell mass. Hematocrit has its own normal range as part of the CBC panel.

#414 · Clinical Hematology

What are RBC indices?

Answer

RBC indices are calculated values that describe the size and hemoglobin content of an average red blood cell, derived from the red cell count, hemoglobin, and hematocrit. They help characterize the type of anemia a patient may have by describing cell size and color. Knowing their normal ranges is part of describing the CBC.

#415 · Clinical Hematology

What does the WBC count measure in a CBC?

Answer

The WBC count measures the total number of white blood cells in a blood sample, which are part of the immune system. It has its own normal range and can rise or fall in response to infection, inflammation, or other conditions. It is one of the core components described in the CBC competency.

#416 · Clinical Hematology

What does the platelet count measure in a CBC?

Answer

The platelet count measures the number of platelets in the blood, which are cell fragments responsible for clot formation. Platelets have their own normal range as part of the CBC panel. Both low and high platelet counts can have clinical significance for bleeding or clotting risk.

#417 · Clinical Hematology

What is the differential in a CBC?

Answer

The differential breaks the total white blood cell count down into the percentages or numbers of each specific white cell type present, such as neutrophils and lymphocytes. It gives more detail than the total WBC count alone. The differential is one of the six core components of the CBC listed in the MLA/T competency profile.

#418 · Clinical Hematology

What does a lipemic hematology sample look like and why does it matter?

Answer

A lipemic sample is cloudy or milky from high fat content in the blood, similar to lipemia in chemistry samples. Lipemia can interfere with automated hematology analyzer readings and needs to be recognized before results are released. This is a hands-on recognition skill using a real specimen, not a simulation-based one.

#419 · Clinical Hematology

Why is a clotted sample a problem for hematology testing?

Answer

A clotted sample means the blood has begun to coagulate in the tube instead of staying anticoagulated, which can consume platelets and cells into the clot and skew cell counts. A clotted CBC sample cannot give an accurate result and typically needs to be redrawn. Recognizing clotted samples is a required hands-on hematology competency.

#420 · Clinical Hematology

Why is an agglutinated sample a problem for hematology testing?

Answer

An agglutinated sample has red blood cells clumping together rather than remaining evenly dispersed, which can cause the analyzer to miscount cells or falsely alter indices like MCV. This clumping can be due to a cold agglutinin or other cause and needs to be recognized before the result is trusted. Staff must take appropriate action, such as warming the sample or repeating the count.

#421 · Clinical Hematology

What is the basic operation of an automated hematology analyzer?

Answer

An automated hematology analyzer counts and characterizes blood cells, usually by passing diluted blood through a detection system that identifies cell size and type electronically or optically. This allows rapid, consistent counting of red cells, white cells, and platelets for the CBC. Understanding this basic operation helps staff recognize normal versus abnormal instrument behavior.

#422 · Clinical Hematology

What is the basic operation of an automated coagulation analyzer?

Answer

An automated coagulation analyzer measures how long it takes a plasma sample to clot after reagents are added, which reflects the function of the coagulation pathway. It is used to run tests like PT and PTT quickly and consistently. Describing this basic operation is part of the hematology competency profile.

#423 · Clinical Hematology

What does PT measure in coagulation testing?

Answer

PT, or prothrombin time, measures how long it takes plasma to clot through the extrinsic coagulation pathway. It is one of the most common coagulation tests and has its own normal range. PT is often used together with INR to monitor patients on certain blood-thinning medications.

#424 · Clinical Hematology

What does PTT measure in coagulation testing?

Answer

PTT, or partial thromboplastin time, measures how long it takes plasma to clot through the intrinsic coagulation pathway. It has its own normal range and is commonly ordered alongside PT to evaluate overall clotting function. PTT is one of the most common coagulation tests an MLA/T should recognize.

#425 · Clinical Hematology

What is INR and how does it relate to PT?

Answer

INR, or international normalized ratio, is a standardized way of reporting PT results so they can be compared consistently across different labs and reagents. It is especially used to monitor patients on warfarin therapy. INR is one of the most common coagulation values an MLA/T should recognize along with its normal range.

#426 · Clinical Hematology

What is a blood film used for?

Answer

A blood film, or blood smear, is a thin layer of blood spread on a slide that allows direct microscopic examination of cell shape, size, and appearance. It can reveal abnormalities that a cell counter alone cannot detect. Preparing blood films correctly is a core hematology competency.

#427 · Clinical Hematology

What is the difference between a thick smear and a thin smear?

Answer

A thick smear uses a larger amount of blood concentrated in a small area, which is useful for detecting low numbers of certain organisms or cells. A thin smear spreads blood in a single layer, which is better for examining individual cell shape and detail. Both types have specific preparation techniques the MLA/T must follow.

#428 · Clinical Hematology

What is a reticulocyte preparation used for?

Answer

A reticulocyte preparation is a specialized smear used to identify and count reticulocytes, which are young, immature red blood cells. Counting reticulocytes helps assess how actively the bone marrow is producing new red cells. Preparing reticulocyte smears is listed alongside blood films and thick and thin smears in the hematology competency.

#429 · Clinical Hematology

Why does the MLA/T need to know both routine and special staining procedures?

Answer

Routine staining, such as a standard blood film stain, is used for everyday CBC and differential review, while special stains highlight specific cell features for less common testing needs. Following the lab's exact staining procedure for each type ensures cells appear correctly under the microscope. This competency covers performing staining according to established lab procedures.

#430 · Clinical Hematology

What is a common staining problem in blood film preparation?

Answer

A common staining problem is uneven or poor staining that leaves cells too pale, too dark, or with a precipitate obscuring detail, often due to timing, pH, or technique issues. The MLA/T must be able to identify these problems by inspecting the slide. Once identified, corrective action such as restaining or adjusting technique should be initiated.

#431 · Clinical Hematology

What is one example of corrective action for a staining problem?

Answer

If a stain appears too dark or too light, the MLA/T may need to adjust the staining time, refresh the stain reagents, or repeat the staining process on a new smear. Identifying the root cause of the problem, such as expired stain or improper pH, guides the correct fix. Initiating corrective action is a required part of this hematology competency.

#432 · Clinical Hematology

What does preparing a body fluid specimen for hematology testing involve?

Answer

Preparing a body fluid specimen involves handling fluids other than whole blood, such as those from joints or body cavities, so they can be properly examined under the microscope. This includes appropriate processing steps before microscopic examination. It is a distinct hematology competency separate from routine blood smear preparation.

#433 · Clinical Hematology

Why do body fluid specimens need special preparation before microscopic exam?

Answer

Body fluids can vary widely in cell concentration, clarity, and composition compared to whole blood, so they often require specific processing like concentration or dilution before an accurate microscopic count can be done. Following the correct preparation steps ensures the cells seen under the microscope reflect the true specimen. This is why body fluid preparation is a distinct hematology competency.

#434 · Clinical Hematology

What is the purpose of a counting chamber in hematology?

Answer

A counting chamber, such as a hemocytometer, is a specialized slide with a grid used to manually count cells under the microscope when an automated count is not suitable or needs to be confirmed. It requires proper dilution of the sample before use. Describing its use, including dilution and flooding the chamber, is a specific hematology competency.

#435 · Clinical Hematology

Why must a sample be diluted before using a counting chamber?

Answer

Whole blood or fluid contains too many cells to count accurately without dilution, so a measured dilution brings the cell concentration down to a countable range. The dilution factor is then used in the final calculation to determine the true cell count. Getting the dilution correct is essential for an accurate manual count.

#436 · Clinical Hematology

What does flooding the counting chamber mean?

Answer

Flooding the counting chamber means applying the diluted sample to the chamber so it evenly fills the counting area under the coverslip without overflowing or leaving air bubbles. Proper flooding ensures cells are evenly distributed across the grid for an accurate count. Poor flooding technique can lead to inaccurate manual cell counts.

#437 · Clinical Hematology

What is the erythrocyte sedimentation rate (ESR) test measuring?

Answer

ESR measures how far red blood cells settle, or sediment, in a vertical tube over a set period of time, usually one hour. A faster sedimentation rate can be associated with inflammation in the body. Setting up ESR testing correctly and knowing potential sources of error is a specific hematology competency.

#438 · Clinical Hematology

Name one potential source of error in ESR testing.

Answer

A common source of error is not keeping the ESR tube perfectly vertical during the timed period, since tilting can cause cells to settle faster and give a falsely elevated result. Other errors can include incorrect timing or an improperly mixed sample. Recognizing these sources of error is part of the ESR setup competency.

#439 · Clinical Hematology

What is the Westergren method for ESR testing?

Answer

The Westergren method is a widely used ESR technique that uses a specific tube length and format to measure how far red cells settle over a set time period. It is one of the two methods the MLA/T must be able to differentiate. Understanding its setup helps ensure accurate, standardized results.

#440 · Clinical Hematology

What is the Wintrobe method for ESR testing?

Answer

The Wintrobe method is another ESR technique that uses a different tube format than the Westergren method to measure red cell sedimentation. The MLA/T must be able to differentiate between the Westergren and Wintrobe methods as part of the hematology competency profile. Each method has its own specific setup and considerations.

#441 · Clinical Hematology

Why is it important to differentiate between Westergren and Wintrobe ESR methods?

Answer

The two methods use different tube designs and setups, so results are not automatically interchangeable between them. Knowing which method is being used ensures the test is set up correctly and helps avoid confusion when interpreting or reporting results. This differentiation is explicitly named as its own hematology competency.

#442 · Clinical Hematology

What does it mean for a hematology test to require special handling?

Answer

Some hematology tests need extra steps beyond routine sample handling, such as specific temperature control, timing, or transport conditions, to give an accurate result. Identifying which tests require special handling helps the MLA/T avoid preanalytical errors. This is listed as its own distinct hematology competency.

#443 · Clinical Hematology

Why must the MLA/T recognize samples that require special handling before processing them normally?

Answer

If a test requiring special handling is processed the same way as a routine sample, the result may be inaccurate or the test may need to be repeated. Recognizing these tests ahead of time allows staff to apply the correct temperature, timing, or transport steps from the start. This protects both the sample and the patient from unnecessary redraws.

#444 · Clinical Hematology

Why can up to 90% of clinical hematology competencies be assessed in simulation?

Answer

Most hematology skills, including describing CBC components, coagulation tests, smear preparation, staining, and ESR setup, can be taught and evaluated through simulated scenarios and written or practical demonstration. Recognizing real lipemic, clotted, or agglutinated samples is the one competency that requires an actual physical specimen. This is why hematology has a higher simulation-eligible percentage than chemistry.

#445 · Clinical Hematology

What is the relationship between hemoglobin and hematocrit results?

Answer

Hemoglobin and hematocrit both reflect the amount of red blood cell mass in the blood and typically rise and fall together. Hematocrit is often roughly three times the hemoglobin value in a normal sample, though this is only a general pattern rather than a fixed rule. Both values are core components of the CBC with their own normal ranges.

#446 · Clinical Hematology

Why is the WBC differential useful beyond the total WBC count?

Answer

The total WBC count only shows how many white cells are present overall, while the differential breaks that number down by specific cell type, showing which types are elevated or decreased. This added detail helps distinguish between different causes of an abnormal WBC count, such as infection type or other conditions. The differential is a required part of describing CBC components.

#447 · Clinical Hematology

What action should be taken with a clotted or agglutinated hematology sample?

Answer

The MLA/T should recognize that the sample is unsuitable for accurate automated counting and take the appropriate action, which may include redrawing the sample or applying a technique like warming for agglutination. Running a clotted or agglutinated sample through the analyzer without addressing the issue can produce a misleading CBC result. This recognition is a required hands-on, non-simulation competency.

#448 · Clinical Hematology

How does PT/INR monitoring relate to patient care?

Answer

PT and INR are commonly used to monitor patients taking blood-thinning medication like warfarin, helping ensure their dose keeps clotting time in a safe therapeutic range. Recognizing normal ranges for these coagulation tests helps staff understand when a result may need urgent clinical attention. This is why PT, PTT, and INR are named as the most common coagulation tests to know.

#449 · Clinical Hematology

Why is proper blood film preparation important for an accurate differential?

Answer

A well-prepared blood film spreads cells in a single, evenly distributed layer so they can be clearly identified and counted under the microscope. A poorly prepared smear can distort cell shape or clump cells together, leading to an inaccurate manual differential. This is why preparing smears correctly is its own listed hematology competency.

#450 · Clinical Hematology

What is the purpose of performing a manual differential using a stained blood film?

Answer

A manual differential lets a trained eye examine and classify white blood cells directly under the microscope, which can catch abnormal cells or patterns that an automated analyzer might flag or miss. It relies on both proper smear preparation and proper staining to be accurate. This connects the staining and smear preparation competencies to real patient care.

#451 · Clinical Hematology

Why is knowing normal ranges for CBC components not enough on its own?

Answer

Knowing the normal range for each component helps identify an abnormal result, but the MLA/T must also understand sample quality issues, like clotting or agglutination, that can cause a falsely abnormal reading. Combining knowledge of normal ranges with sample recognition skills ensures results reflect the patient's true status. This is why both areas are separate competencies within clinical hematology.

#452 · Clinical Hematology

What is one reason a coagulation sample might need special handling?

Answer

Coagulation tests like PT and PTT can be affected by improper sample volume, tube type, or delayed processing, since the coagulation pathway is sensitive to these preanalytical factors. Recognizing coagulation testing as needing special handling helps prevent inaccurate PT, PTT, or INR results. This ties into the broader competency of identifying hematology tests that require special handling.

#453 · Clinical Hematology

How do thick and thin smears complement each other in hematology?

Answer

A thick smear concentrates blood to help detect low numbers of certain organisms or cells that might otherwise be missed, while a thin smear preserves individual cell shape for detailed morphology review. Using both together can give a more complete picture than either alone. Preparing both types correctly is part of the hematology smear competency.

#454 · Clinical Hematology

Why is reticulocyte counting clinically useful?

Answer

Reticulocytes are immature red blood cells, so counting them gives an indication of how actively the bone marrow is producing new red cells in response to conditions like anemia or blood loss. A low reticulocyte count in an anemic patient can suggest a bone marrow production problem, while a high count suggests an appropriate marrow response. This is why reticulocyte preparation is listed alongside other core smear techniques.

#455 · Clinical Hematology

Summarize why sample quality recognition, covering lipemic, clotted, and agglutinated samples, is a core clinical hematology competency.

Answer

These sample defects can distort automated cell counts and coagulation results in specific, predictable ways, so recognizing them protects the accuracy of the CBC and related tests. This competency requires hands-on judgment with real specimens rather than only simulation. It directly connects specimen handling skills to reliable patient results in hematology.

#456 · Transfusion Medicine

What are the four main ABO blood groups?

Answer

The four groups are A, B, AB, and O, based on which antigens sit on the surface of red blood cells. Group A has A antigens, group B has B antigens, group AB has both, and group O has neither. Each person's plasma also carries antibodies against whatever antigen they lack, which is why matching blood type matters so much for safe transfusion.

#457 · Transfusion Medicine

Why is group O negative blood called the universal donor?

Answer

O negative red cells carry no A, B, or Rh(D) antigens, so they are far less likely to trigger an antibody reaction in a recipient of any blood type. This makes O negative units the safest emergency choice when there is no time to determine a patient's exact type. It is still not perfectly universal, since other minor antigens can still cause reactions in rare cases.

#458 · Transfusion Medicine

What does Rh typing determine and why does it matter?

Answer

Rh typing checks whether the Rh(D) antigen is present on a patient's red blood cells, making them Rh positive or Rh negative. An Rh negative person exposed to Rh positive blood can develop antibodies against it, which is dangerous for future transfusions and for an Rh negative mother carrying an Rh positive baby. This is why both ABO and Rh status are confirmed before any transfusion.

#459 · Transfusion Medicine

What is antibody screening used for in transfusion medicine?

Answer

Antibody screening tests a patient's plasma against a panel of reagent red cells to detect unexpected antibodies beyond the basic ABO and Rh antibodies. A positive result means further identification testing is needed before compatible blood can be issued safely. This step helps prevent delayed or acute hemolytic transfusion reactions from antibodies the patient developed from prior transfusions or pregnancies.

#460 · Transfusion Medicine

What anticoagulant is typically used to collect a specimen for blood group and crossmatch testing?

Answer

Transfusion medicine samples are usually collected in a pink or lavender top tube containing EDTA, the same anticoagulant used for hematology testing. EDTA keeps the red cells intact and prevents clotting so ABO/Rh typing and antibody testing can be performed accurately. Using the wrong tube type can compromise results and may require the sample to be rejected and redrawn.

#461 · Transfusion Medicine

Name three reasons a transfusion medicine specimen might be rejected.

Answer

A sample can be rejected if it is unlabeled or mislabeled, if the patient identification on the tube does not exactly match the requisition, or if the sample is clotted, hemolyzed, or in the wrong tube type. Because a mismatched transfusion can be fatal, transfusion medicine labs enforce stricter labeling and identification rules than almost any other lab section. When in doubt, the safe practice is always to reject and recollect rather than guess.

#462 · Transfusion Medicine

What is packed red blood cells (PRBCs) and what is it used for?

Answer

Packed red blood cells is whole blood with most of the plasma removed, leaving a concentrated unit of red cells used to treat anemia or blood loss. It restores oxygen-carrying capacity without giving the patient unnecessary extra fluid volume. PRBCs are typically stored refrigerated and have a shelf life of about 42 days depending on the additive solution used.

#463 · Transfusion Medicine

How is fresh frozen plasma (FFP) stored and what is it used for?

Answer

FFP is the liquid portion of blood separated and frozen soon after collection to preserve clotting factors, and it must be kept frozen until it is thawed for use. It is given to patients with clotting factor deficiencies or active bleeding related to coagulation problems. Once thawed, it has a limited window during which it must be transfused before the factors degrade.

#464 · Transfusion Medicine

What is cryoprecipitate and how is it obtained?

Answer

Cryoprecipitate is prepared by thawing fresh frozen plasma slowly at a cold temperature and collecting the precipitate that forms, which is rich in fibrinogen and clotting factor VIII. It is used for patients with fibrinogen deficiency or certain bleeding disorders. Like other plasma products, it must be handled and stored under strict temperature control to preserve its clotting activity.

#465 · Transfusion Medicine

What storage temperature is required for platelet concentrates and why does it differ from red cells?

Answer

Platelets are stored at room temperature, around 20 to 24 degrees Celsius, with continuous gentle agitation rather than being refrigerated. Cold temperatures damage platelet function, so refrigeration used for red cells would make platelets ineffective. Because room temperature storage also supports bacterial growth, platelets have a much shorter shelf life than red cells, typically only a few days.

#466 · Transfusion Medicine

What is stock rotation and why is it important for blood products?

Answer

Stock rotation means using the oldest compatible units first, often called first-in-first-out, so that blood products are issued before they expire. This reduces wastage of a limited and perishable resource and ensures patients receive products that are still within their approved shelf life. Expiration dates are checked at every step, from receiving to storage to issue.

#467 · Transfusion Medicine

What tests are routinely performed on all donated blood units?

Answer

Donated blood is tested for ABO and Rh type, and screened for transfusion transmissible infections such as HIV, hepatitis B and C, syphilis, and other pathogens depending on regional protocols. This screening protects both the recipient and the safety of the blood supply overall. Only units that pass all required testing are released for transfusion use.

#468 · Transfusion Medicine

What anticoagulant is used when collecting whole blood from a donor, and why?

Answer

Whole blood donations are collected into bags containing citrate-based anticoagulant preservative solutions, such as CPD or CPDA-1. Citrate works by binding calcium, which prevents the clotting cascade from proceeding. These solutions also contain nutrients that help keep red cells viable for storage after collection.

#469 · Transfusion Medicine

After whole blood is collected from a donor, how are the components typically prepared?

Answer

Whole blood is centrifuged to separate it into its major components, red blood cells, plasma, and platelets, which are then processed and stored separately under their own specific conditions. This component separation lets one donation help multiple patients, each receiving only the specific component they need. Careful handling during separation helps preserve the quality of each resulting product.

#470 · Transfusion Medicine

Why does an ABO grouping or Rh typing error carry such serious consequences?

Answer

Transfusing a patient with mismatched ABO blood can trigger a severe acute hemolytic reaction, where the immune system rapidly destroys the transfused red cells, which can be life threatening. Because the risk is so high, transfusion medicine relies on multiple independent checks, including patient identification, sample labeling, and crossmatching, before any unit is released. This is why the transfusion medicine section holds some of the strictest error-prevention rules in the entire lab.

#471 · Electrocardiograms and Holter Monitors

What is the natural pacemaker of the heart?

Answer

The sinoatrial (SA) node, located in the wall of the right atrium, is the heart's natural pacemaker because it normally initiates each heartbeat. It generates an electrical impulse that spreads across both atria, causing them to contract. This impulse then travels toward the atrioventricular node to continue through the conduction system.

#472 · Electrocardiograms and Holter Monitors

What is the role of the atrioventricular (AV) node?

Answer

The AV node receives the electrical impulse from the atria and briefly delays it before passing it on to the ventricles. This delay allows the atria to finish contracting and empty their blood into the ventricles before the ventricles contract. From the AV node, the impulse continues down into the bundle of His.

#473 · Electrocardiograms and Holter Monitors

Describe the pathway of the cardiac conduction system after the AV node.

Answer

After the AV node, the impulse travels down the bundle of His, which splits into the right and left bundle branches running along the septum between the ventricles. From there it spreads into the Purkinje fibers, which fan out through the ventricular walls. This rapid spread causes both ventricles to contract together in a coordinated pump.

#474 · Electrocardiograms and Holter Monitors

What does the P wave on an ECG tracing represent?

Answer

The P wave represents the electrical activity of atrial depolarization, meaning the electrical signal spreading across the atria just before they contract. It is normally a small, rounded upward deflection preceding each QRS complex. An abnormal or missing P wave can point to problems with the SA node or atrial rhythm.

#475 · Electrocardiograms and Holter Monitors

What does the QRS complex represent on an ECG?

Answer

The QRS complex represents ventricular depolarization, the electrical signal spreading through the ventricles that triggers their contraction. It is normally the largest, sharpest deflection on the tracing because the ventricles have much more muscle mass than the atria. Its shape, width, and timing are closely examined to detect conduction problems.

#476 · Electrocardiograms and Holter Monitors

What does the T wave represent on an ECG?

Answer

The T wave represents ventricular repolarization, the phase where the ventricles' electrical activity resets after contracting so they are ready for the next beat. It normally appears as a smooth rounded wave following the QRS complex. Changes in the shape or direction of the T wave can indicate issues such as ischemia or electrolyte imbalance.

#477 · Electrocardiograms and Holter Monitors

How many leads make up a standard 12-lead ECG and what are the two main categories?

Answer

A standard 12-lead ECG is made up of 6 limb leads and 6 precordial (chest) leads. The limb leads look at the heart's electrical activity from the frontal plane using electrodes on the arms and legs, while the precordial leads look at it from the horizontal plane using electrodes placed across the chest. Together they give a complete picture of the heart's electrical activity from multiple angles.

#478 · Electrocardiograms and Holter Monitors

Where is electrode V1 placed for a 12-lead ECG?

Answer

V1 is placed in the fourth intercostal space, just to the right of the sternum. It is one of the two starting reference points for placing the rest of the precordial leads. Correct placement of V1 is important because an error here throws off the placement of V2 through V6 as well.

#479 · Electrocardiograms and Holter Monitors

Where is electrode V2 placed, and how does it relate to V1?

Answer

V2 is placed in the fourth intercostal space, just to the left of the sternum, directly across from V1. Together V1 and V2 sit at the same horizontal level, framing the sternum on each side. V4 is then placed one rib space down, in the fifth intercostal space, using V1 and V2 as the starting landmark.

#480 · Electrocardiograms and Holter Monitors

Where is electrode V4 placed relative to V1 and V2?

Answer

V4 is placed in the fifth intercostal space at the midclavicular line, which is found by moving down one rib space from the V1 to V2 level. V4 is positioned before V3, since V3 is then placed directly between V2 and V4. This landmark-based order helps keep chest lead placement consistent and accurate.

#481 · Electrocardiograms and Holter Monitors

Where are electrodes V5 and V6 placed?

Answer

V5 is placed at the anterior axillary line, level with V4, and V6 is placed at the midaxillary line, also level with V4. All three of V4, V5, and V6 sit along the same horizontal line around the side of the chest. Keeping them level is important for accurate comparison of the heart's electrical activity across that region.

#482 · Electrocardiograms and Holter Monitors

Where are the four limb electrodes placed for a standard 12-lead ECG?

Answer

The four limb electrodes go on the right arm, left arm, right leg, and left leg, usually on the fleshy part of the forearm and lower leg rather than directly over bone or a joint. The right leg electrode acts as a ground and does not contribute to the tracing itself. Consistent limb placement matters because moving an electrode to the torso can change the tracing's appearance.

#483 · Electrocardiograms and Holter Monitors

What is the standard paper speed and voltage calibration for a routine ECG?

Answer

A standard ECG is typically recorded at a paper speed of 25 millimetres per second with a calibration of 10 millimetres per millivolt. These standard settings let clinicians measure intervals and amplitudes consistently across different machines and patients. If the speed or calibration is changed, it must be clearly noted on the tracing so it is not misread.

#484 · Electrocardiograms and Holter Monitors

What is a Holter monitor and how is it different from a standard resting ECG?

Answer

A Holter monitor is a small portable device that continuously records a patient's heart rhythm, usually over 24 to 48 hours, while they go about normal daily activities. Unlike a standard resting ECG that captures only a brief snapshot, a Holter monitor can catch intermittent arrhythmias that might not show up during a short office visit. Patients often keep a diary of symptoms and activities to correlate with the recorded rhythm.

#485 · Electrocardiograms and Holter Monitors

How does an event monitor differ from a Holter monitor?

Answer

An event monitor is worn for a longer period, often weeks, but it only records when the patient activates it or when it detects an abnormal rhythm, rather than continuously recording everything. This makes it useful for catching rare symptoms that a short-term Holter monitor might miss. A Holter monitor, in contrast, records continuously for its whole wear period regardless of symptoms.

#486 · Electrocardiograms and Holter Monitors

Why must skin be properly prepared before applying ECG electrodes?

Answer

Skin should be cleaned of oils and lotions, and excess hair should be clipped or shaved from the electrode sites so the electrodes make good contact. Poor skin contact increases electrical resistance and leads to a noisy or unreadable tracing. In some cases light abrasion of the skin surface is also used to further improve conductivity.

#487 · Electrocardiograms and Holter Monitors

What is wandering baseline artifact and what commonly causes it?

Answer

Wandering baseline artifact appears as the ECG tracing drifting up and down rather than sitting on a steady line. It is commonly caused by patient movement, breathing, or poorly attached electrodes that shift against the skin. Repositioning or replacing a loose electrode and asking the patient to stay still can usually resolve it.

#488 · Electrocardiograms and Holter Monitors

What causes somatic tremor (muscle) artifact on an ECG and how can it be reduced?

Answer

Somatic tremor artifact appears as fine, irregular spikes on the tracing and is caused by patient muscle movement, shivering, or tension rather than the heart's electrical activity. It can be reduced by making sure the patient is warm, relaxed, and positioned comfortably with limbs well supported. Talking to an anxious patient to help them relax can also reduce this type of interference.

#489 · Electrocardiograms and Holter Monitors

What causes 60-cycle (AC) interference on an ECG tracing?

Answer

Sixty-cycle interference shows up as a regular, fuzzy pattern of small evenly spaced spikes and is caused by nearby electrical equipment or improperly grounded wiring picking up alternating current. Moving the ECG cables and machine away from other electrical devices, and checking that all equipment is properly grounded, often reduces it. Loose electrodes can also make a patient more susceptible to this kind of interference.

#490 · Electrocardiograms and Holter Monitors

What artifact pattern suggests a loose or poorly attached electrode?

Answer

A loose electrode often produces a sudden break, a flat line, or an erratic spike in just one lead while the others remain normal. This points directly to a problem at that specific electrode site rather than a general interference issue. Checking and reattaching the electrode, and reapplying conductive gel if needed, usually fixes it.

#491 · Electrocardiograms and Holter Monitors

What steps should be taken after recording an ECG tracing to assess its quality?

Answer

The technician should review the tracing for a clear steady baseline, adequate amplitude, and freedom from artifact in all 12 leads before ending the procedure. If artifact or a poor-quality section is found, the cause should be identified and corrected, and the affected leads rerecorded. Only a tracing that is clean and technically adequate should be sent on for interpretation.

#492 · Electrocardiograms and Holter Monitors

How should ECG preparation and technique be adapted for pediatric patients?

Answer

Pediatric patients often need smaller electrodes and may require extra reassurance or distraction techniques to help them stay calm and still during the procedure. Their faster resting heart rate compared to adults is a normal finding and should not be mistaken for an abnormality. A caregiver's presence can also help reduce patient anxiety and movement artifact.

#493 · Electrocardiograms and Holter Monitors

How should ECG preparation be adapted for a female patient during precordial electrode placement?

Answer

For female patients, precordial electrodes are placed underneath breast tissue rather than on top of it, so the electrode sits at the correct anatomical landmark on the chest wall. This ensures accurate readings, since placing an electrode on top of breast tissue can distort the tracing. Maintaining patient privacy and comfort during this part of the procedure is also important.

#494 · Electrocardiograms and Holter Monitors

What preventive maintenance and quality control steps help keep ECG equipment ready for use?

Answer

Routine maintenance includes inspecting cables and electrodes for damage, testing the machine against a known calibration standard, and keeping the equipment clean and stored properly between uses. Batteries should be checked and paper supply kept stocked so the machine does not fail mid-procedure. These checks are documented regularly to show the equipment is consistently functioning correctly.

#495 · Electrocardiograms and Holter Monitors

Why is confirming correct patient identification especially important before an ECG or Holter monitor procedure?

Answer

Mislabeling a tracing with the wrong patient's information can lead to a misdiagnosis or an inappropriate treatment decision based on someone else's cardiac data. The technician should verify the patient's identity using at least two identifiers before starting the procedure. The date, time, and patient details must also be accurately recorded on the tracing itself.

#496 · Electrocardiograms and Holter Monitors

What information should a patient record in a Holter monitor diary?

Answer

Patients are asked to log the time and nature of any symptoms they feel, such as dizziness, palpitations, or chest discomfort, along with their activities at that time, such as sleeping, exercising, or resting. This diary lets the interpreting clinician match the recorded rhythm at that exact moment to what the patient was actually experiencing. Without this diary, an abnormal rhythm on the recording is much harder to connect to a real symptom.

#497 · Electrocardiograms and Holter Monitors

Why should a patient avoid showering or bathing while wearing a Holter monitor?

Answer

The Holter monitor and its electrodes are not waterproof, so getting them wet can damage the recording device or cause the electrodes to lose contact with the skin. Patients are instructed to keep the unit dry for the entire wear period, typically 24 to 48 hours. They are also told to avoid strong magnets and certain electrical appliances that could interfere with the recording.

#498 · Electrocardiograms and Holter Monitors

What is the purpose of a right-sided ECG and how is it different from a standard tracing?

Answer

A right-sided ECG places the precordial electrodes on the right side of the chest in a mirror-image pattern of the standard left-sided placement, and is used to evaluate the right ventricle, most often when a right ventricular infarction is suspected. The limb leads are placed the same way as a standard ECG. The resulting tracing should be clearly labeled as a right-sided recording so it is not mistaken for or compared against a standard left-sided one.

#499 · Electrocardiograms and Holter Monitors

What basic electrical safety precautions should be followed when performing an ECG?

Answer

The ECG machine and any related equipment should be properly grounded and inspected for frayed cords or damaged connectors before use around the patient. Wet skin, spilled fluids, or damaged equipment can increase the risk of electrical shock to both patient and technician. Any equipment that appears faulty should be taken out of service and reported rather than used on a patient.

#500 · Electrocardiograms and Holter Monitors

What is the general difference between a resting 12-lead ECG, a Holter monitor, and a stress test as cardiac monitoring tools?

Answer

A resting 12-lead ECG captures a brief snapshot of the heart's electrical activity while the patient is still, a Holter monitor records continuously over one to two days during normal daily activity, and a stress test records the heart's activity while the patient exercises to see how it responds under exertion. Each tool is chosen based on what kind of cardiac problem is suspected, whether it is a baseline abnormality, an intermittent rhythm issue, or exertion-related symptoms. Understanding when each type is appropriate is part of adapting monitoring to the patient's clinical situation.